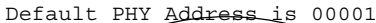


USB TYPE A CONNECTOR

USB TYPE B CONNECTOR

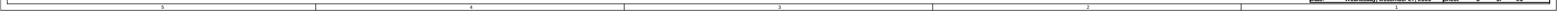
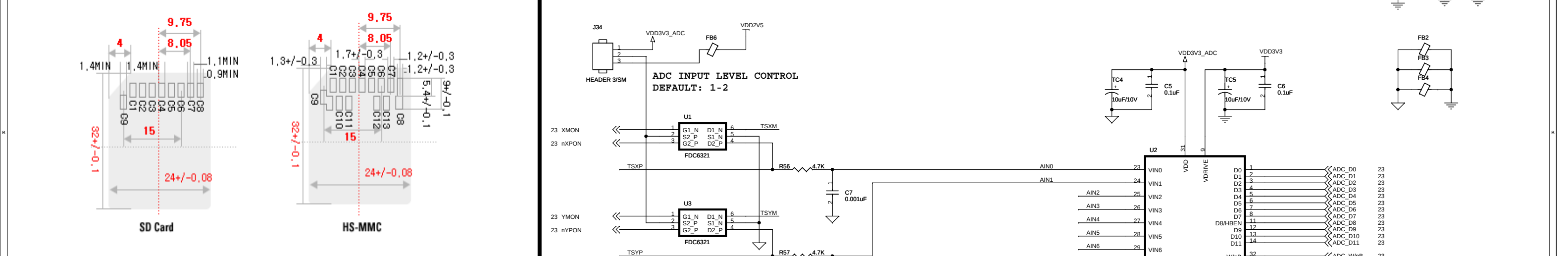
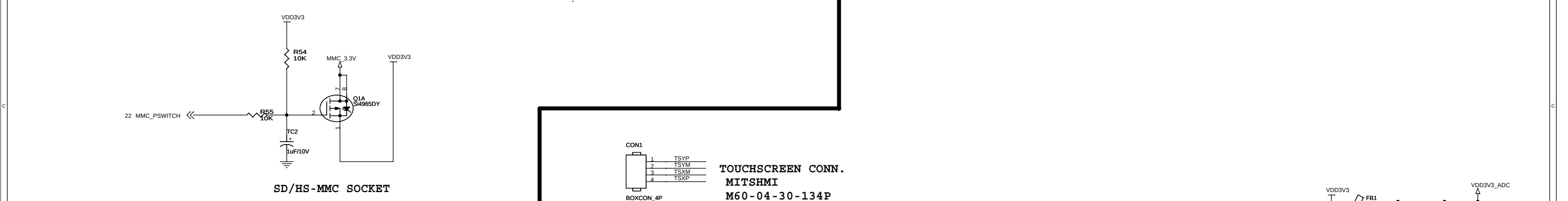
480MHZ DIFFERENTIAL PAIR

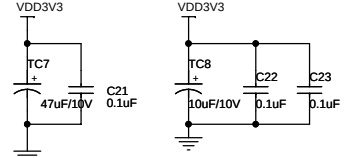
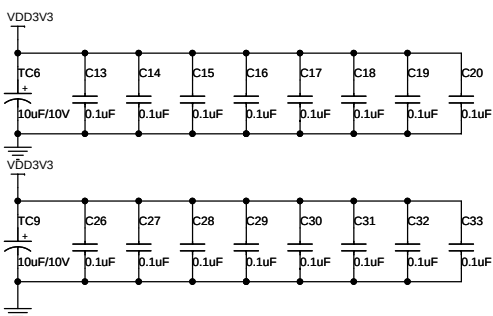
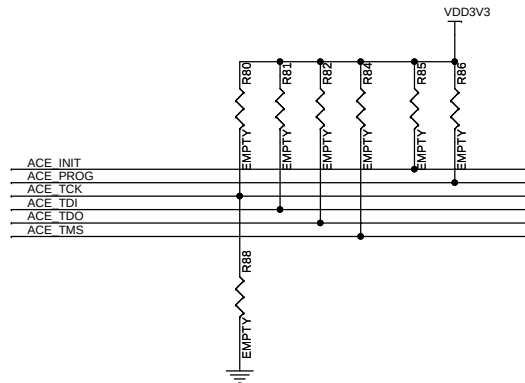
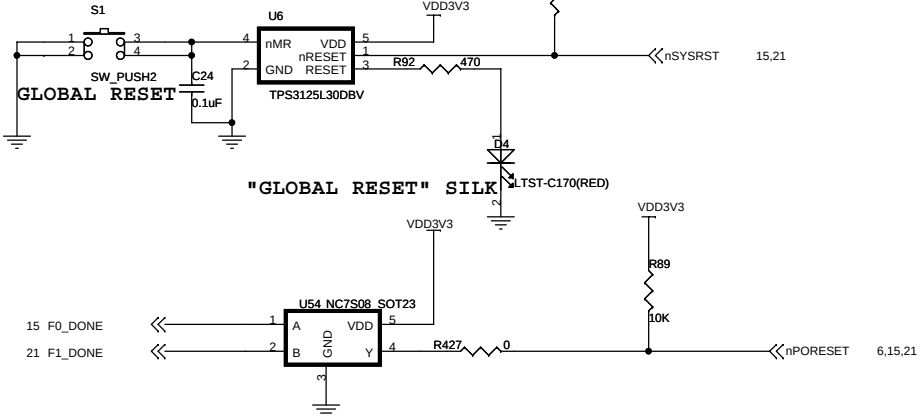
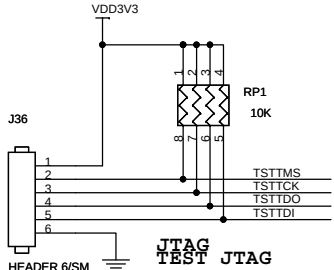
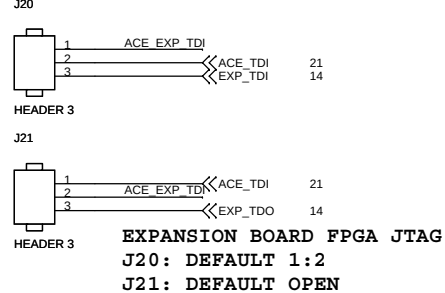
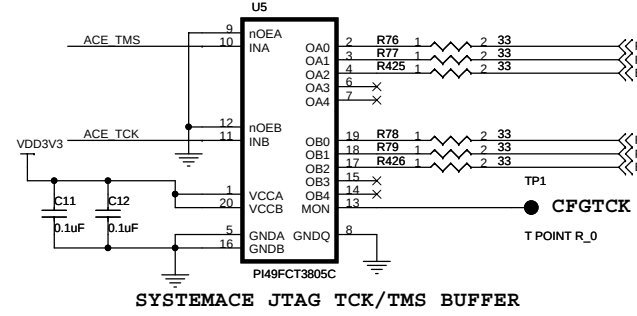
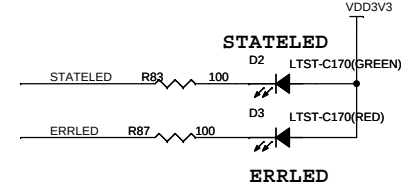
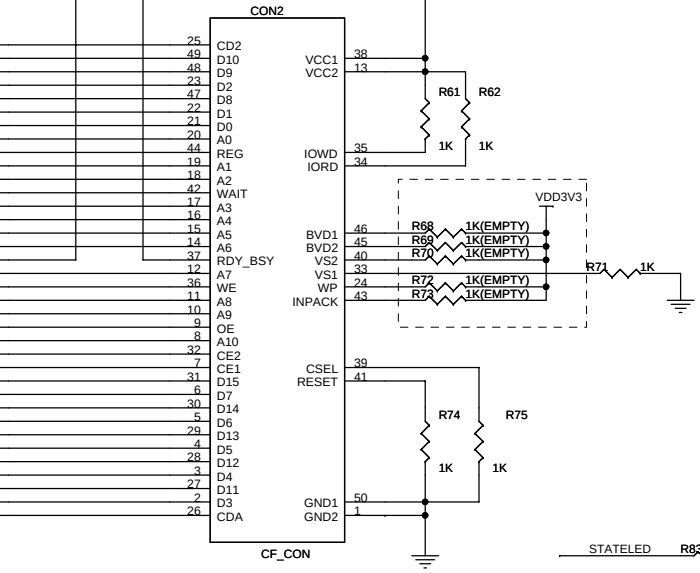
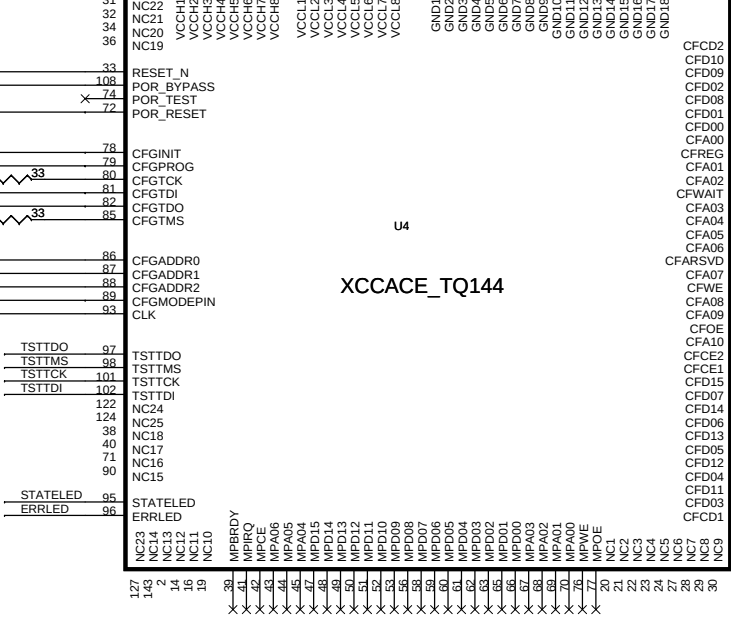
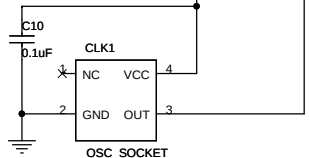
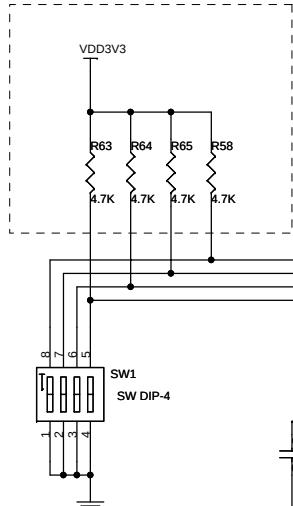
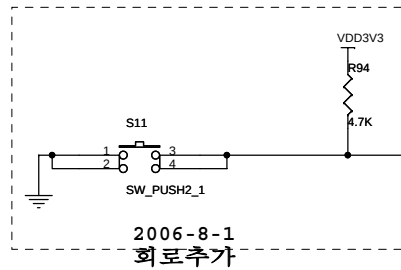


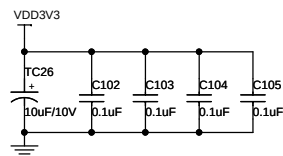
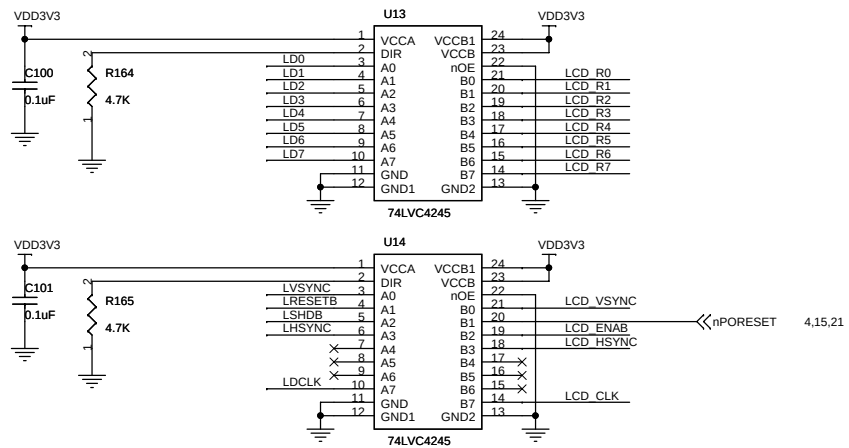
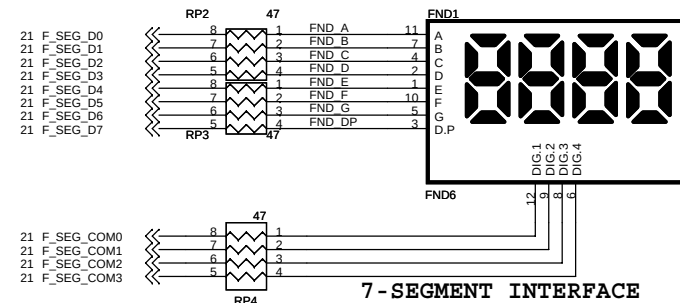
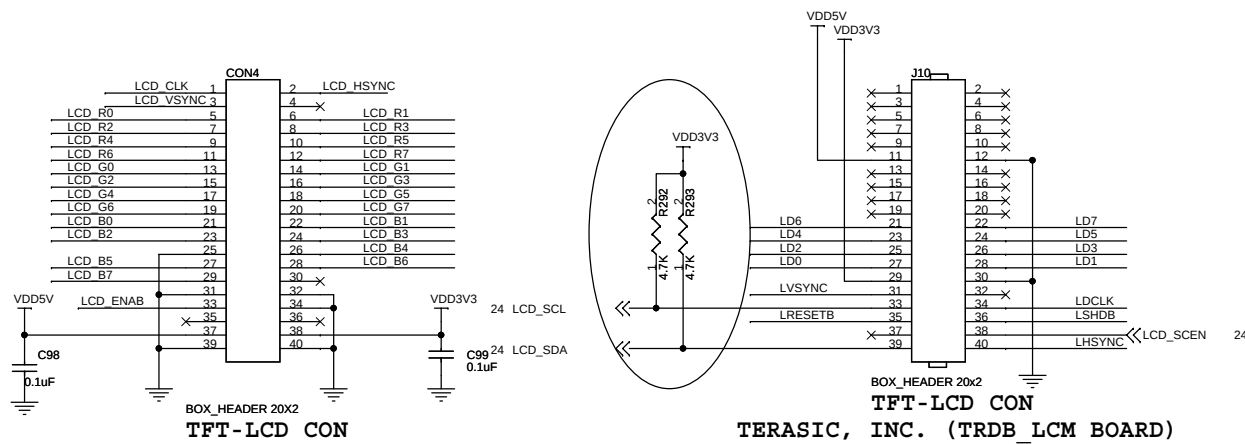
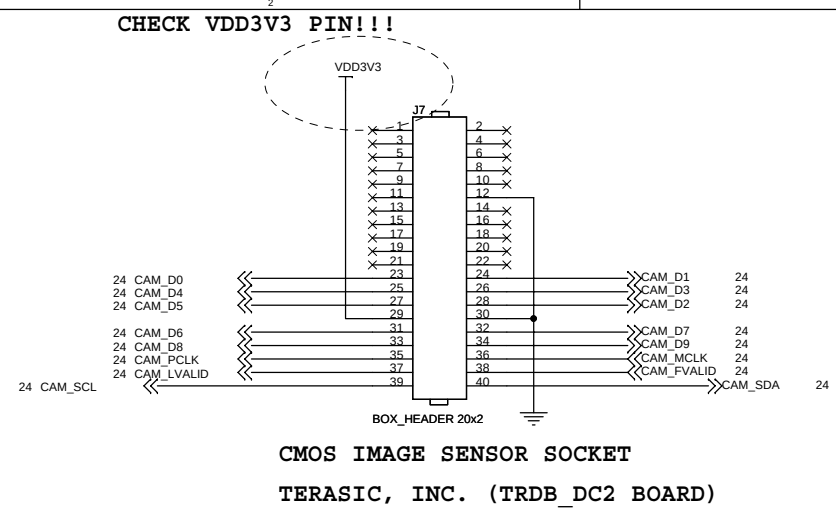
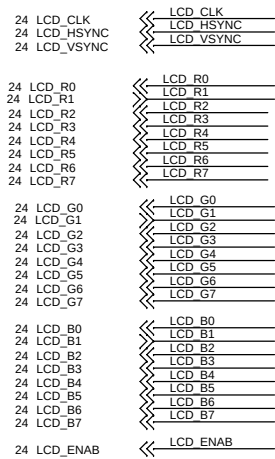
MII ETHERNET PHY

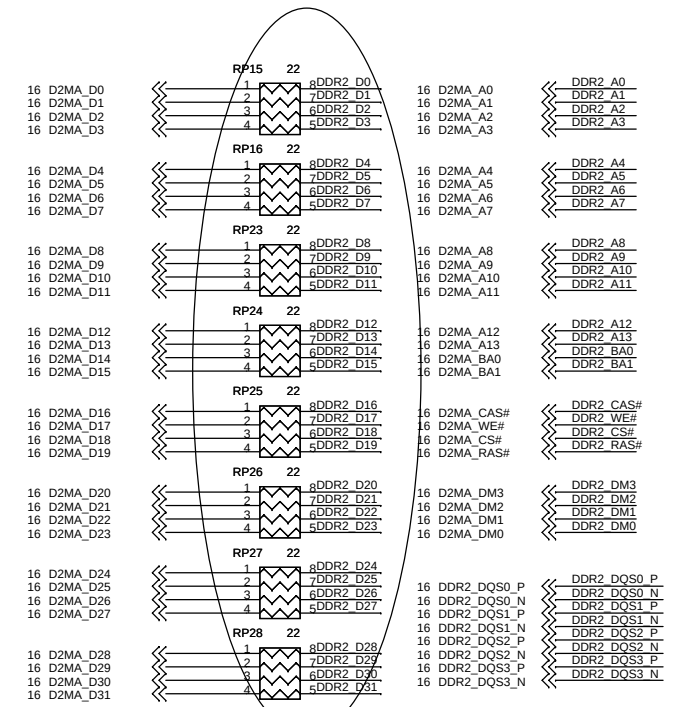
T1 센터에 위치

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Date:	Wednesday, December 27, 2006		Sheet	1 of 1









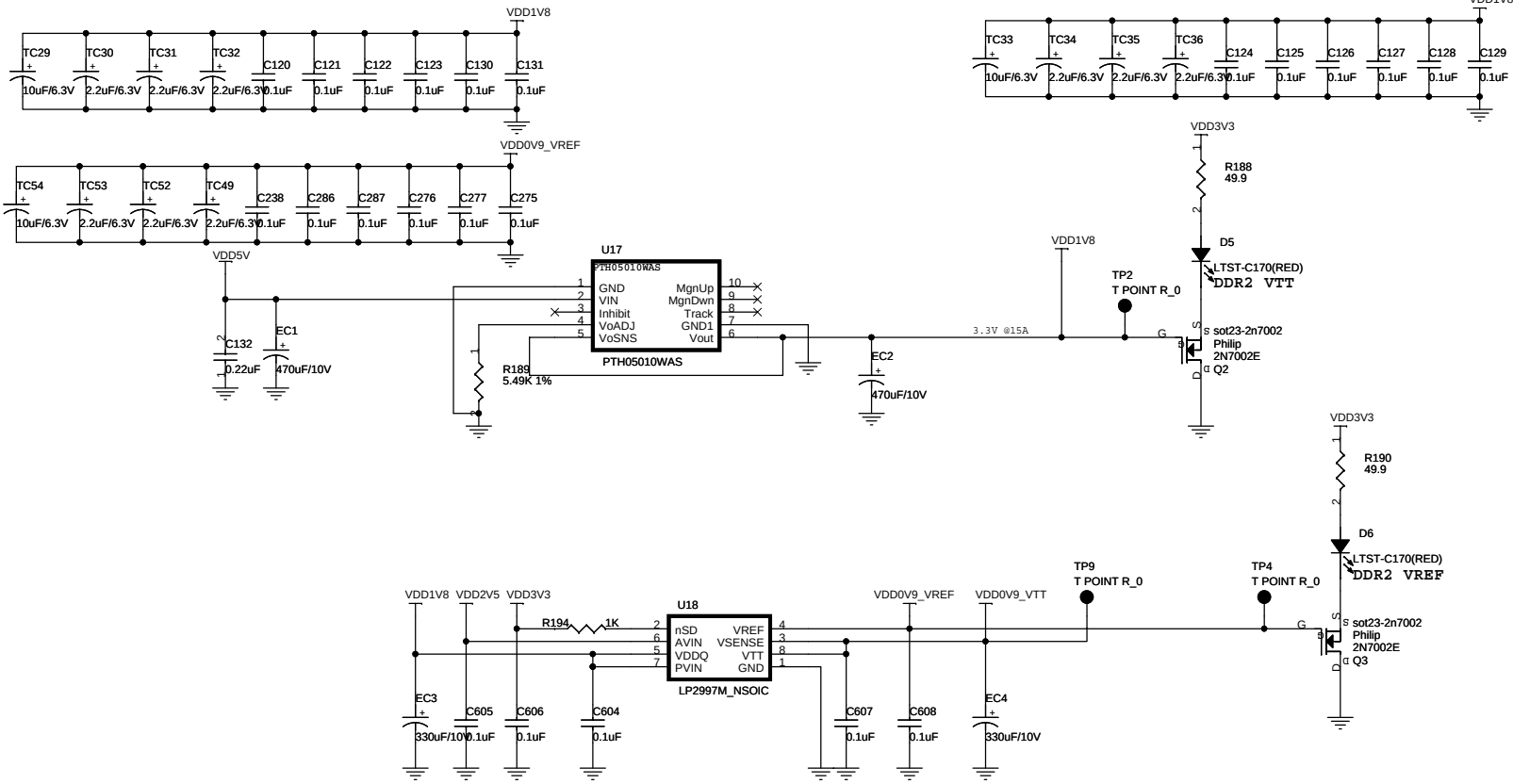
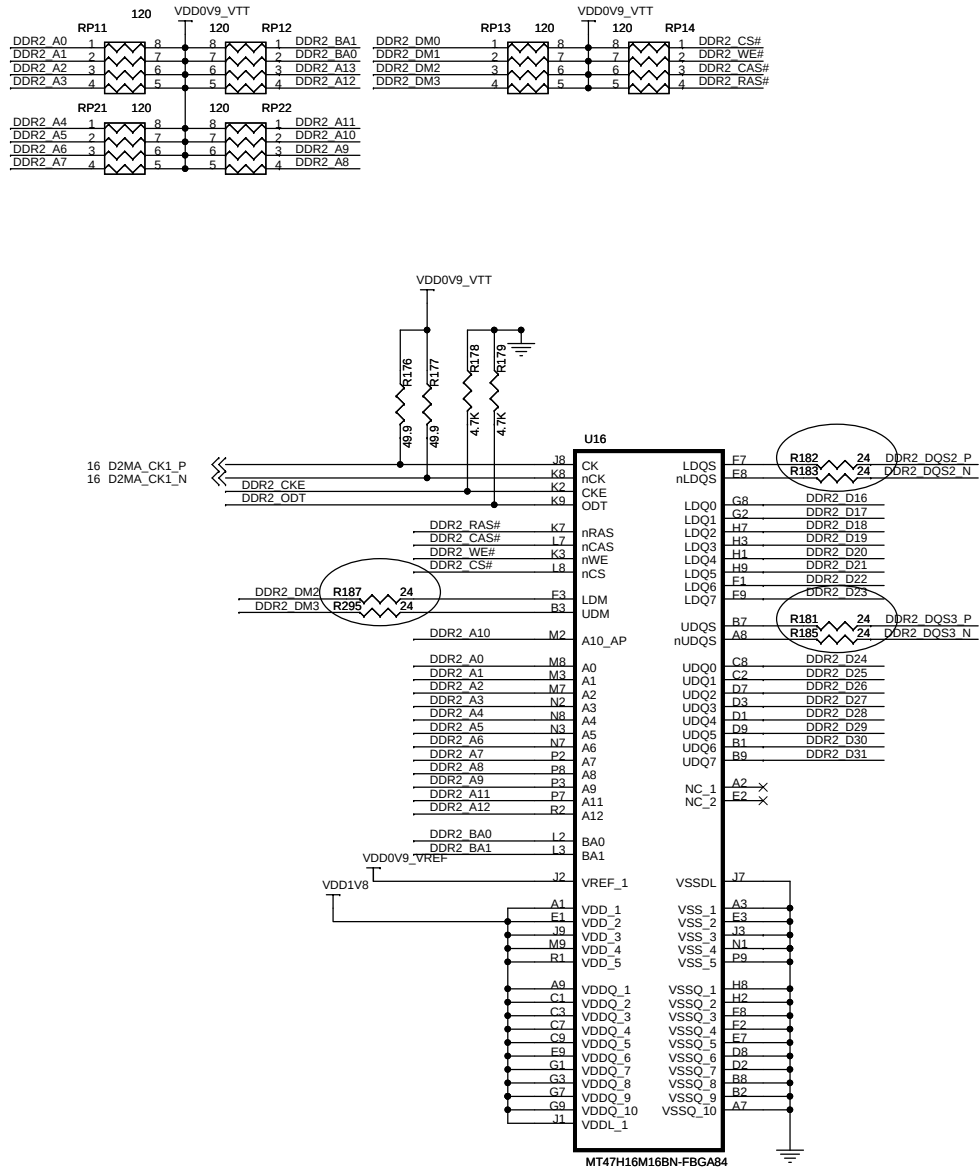
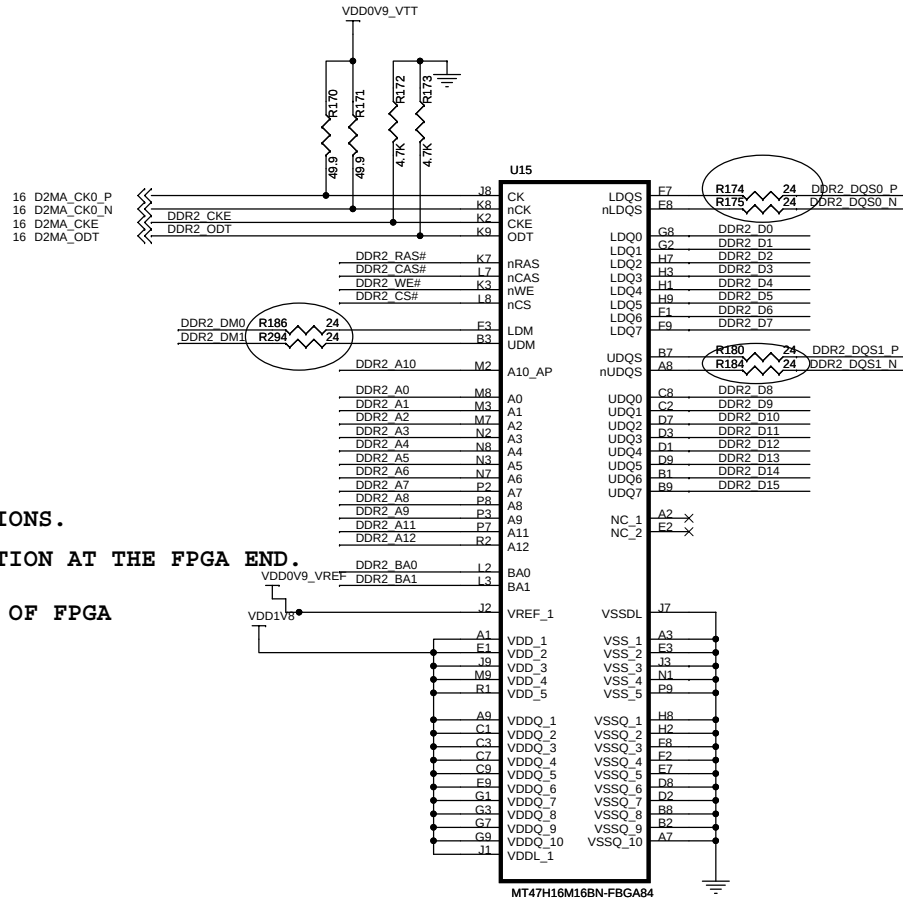
DDR2 SDRAM 에 가깝게 배치

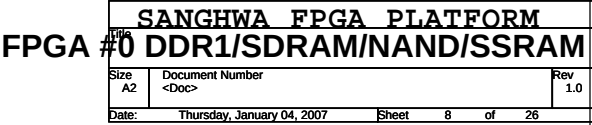
PLACE CLOSE TO DDR MEMORY DEVICES

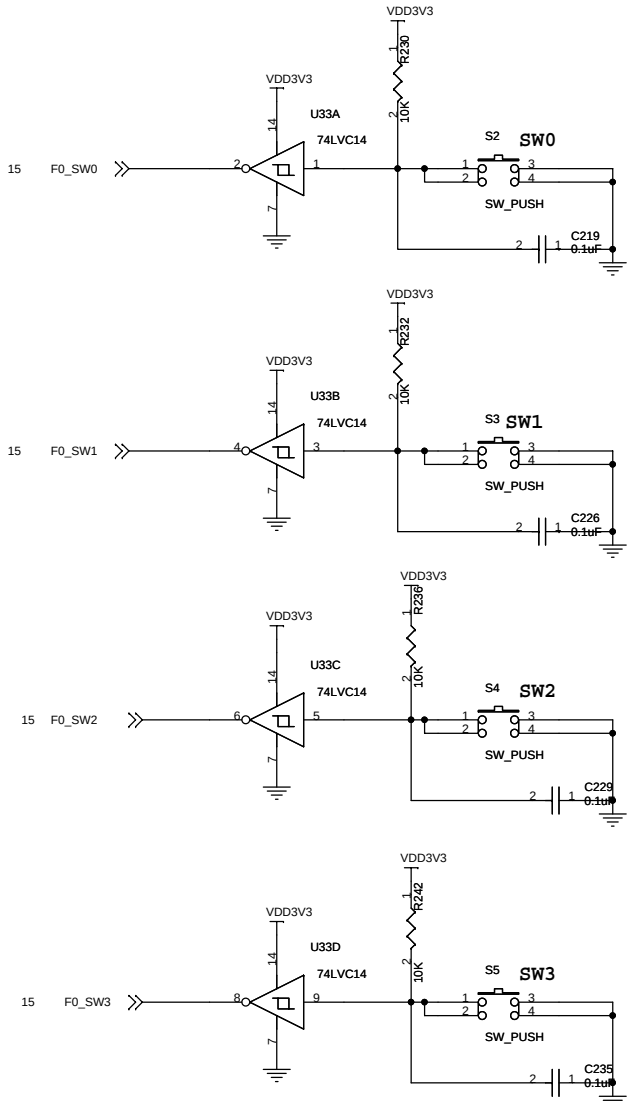
NOTE THAT CK & DQS HAVE DIFFERENTIAL TERMINATIONS.

DQ/DQS FOR COMPONENTS DO NOT HAVE ANY TERMINATION AT THE FPGA END.

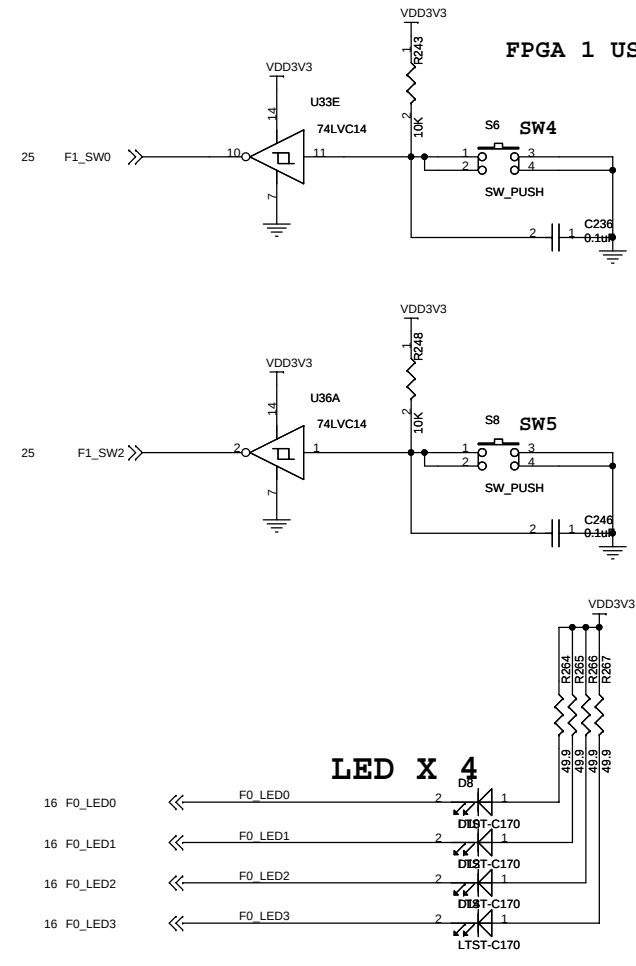
THE INTENTION IS TO RELY ON SSTL18_II DCI I/O OF FPGA FOR TERMINATION.



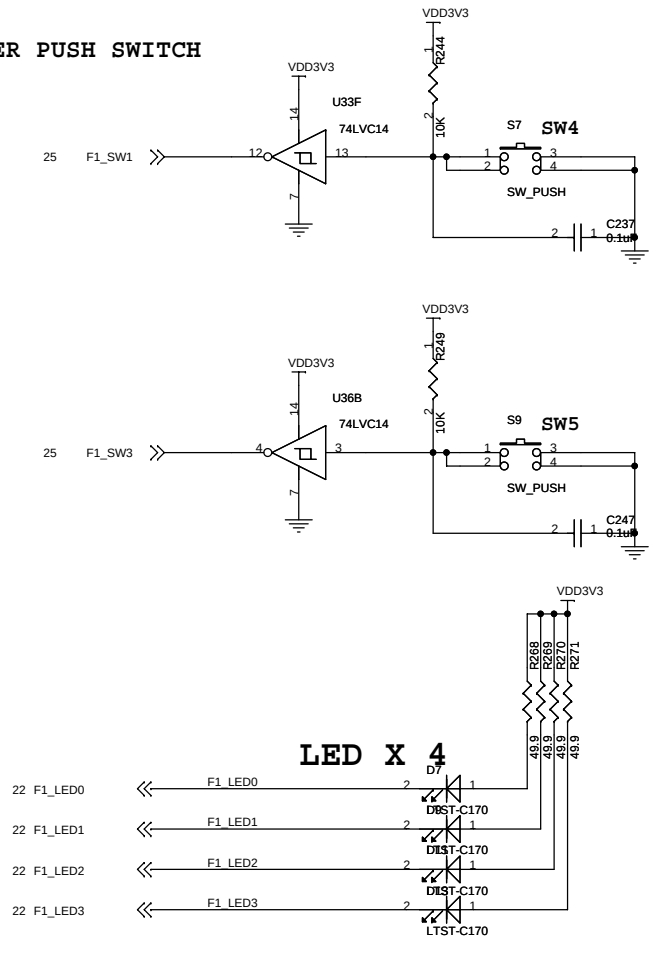




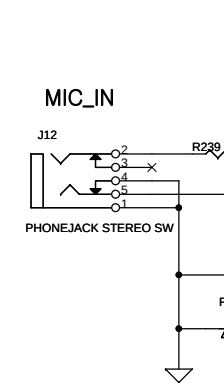
FPGA 0 USER PUSH SWITCH



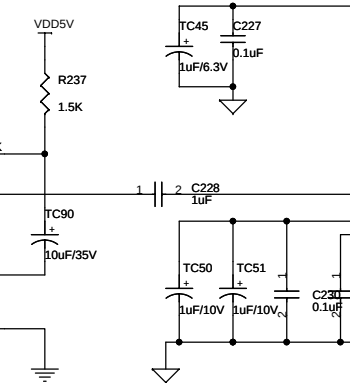
FPGA 1 USER PUSH SWITCH



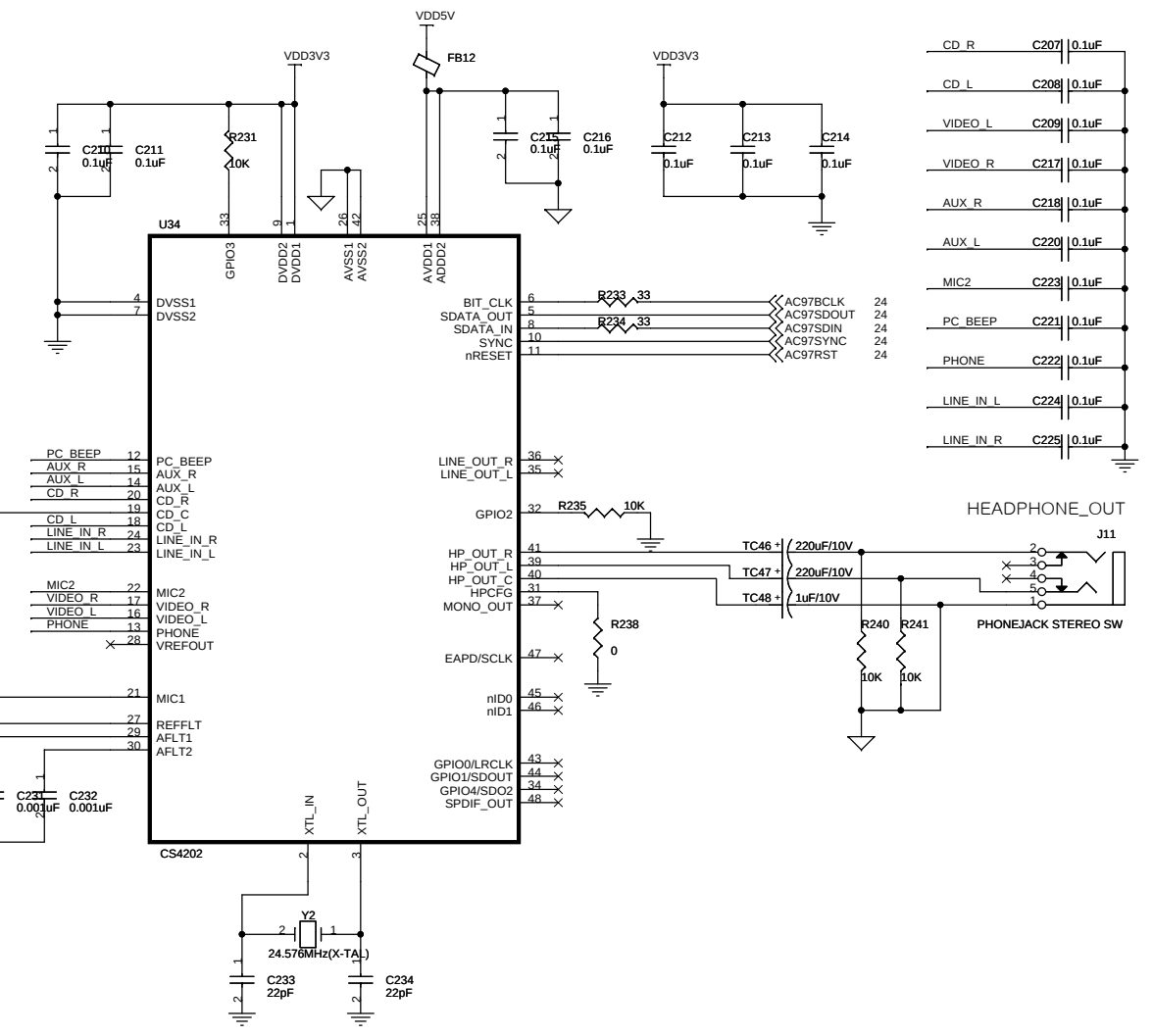
LED X 4



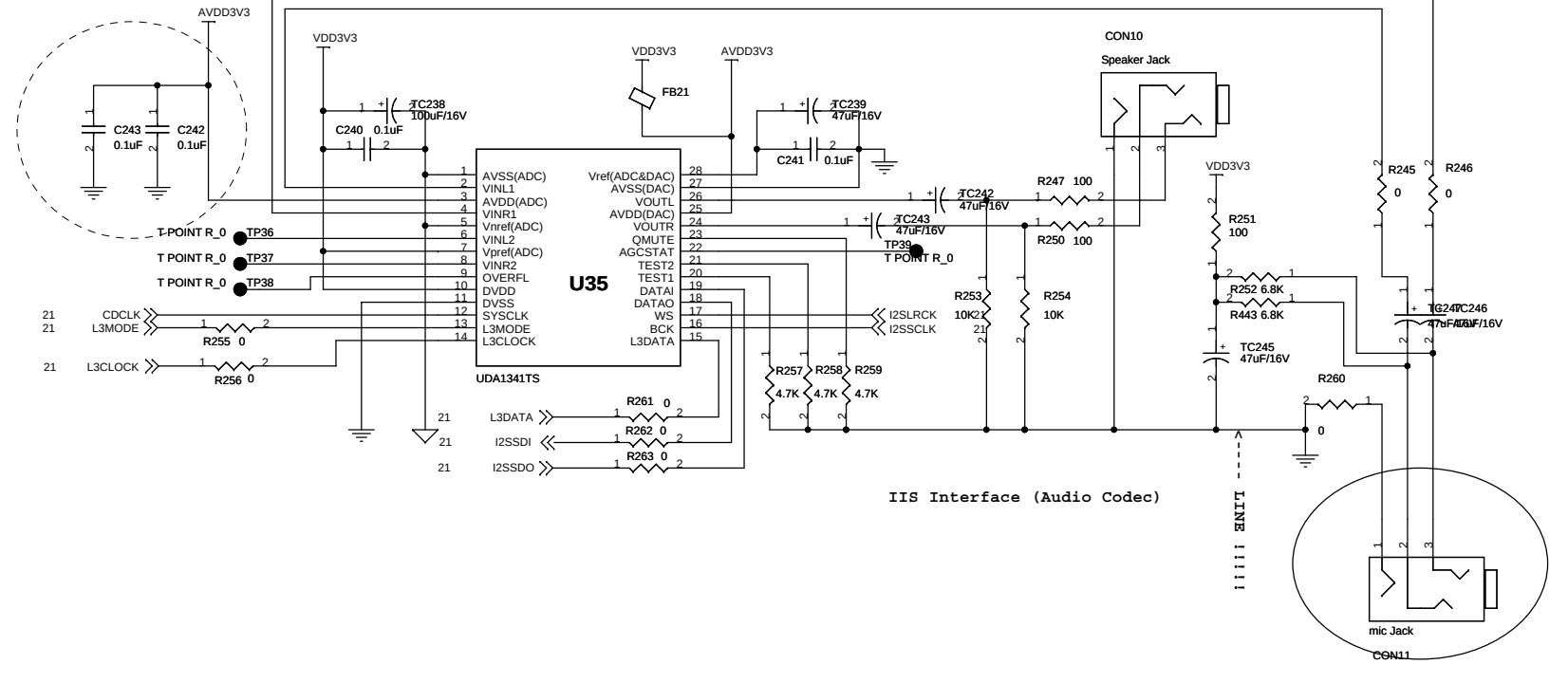
MIC_IN



CS4202



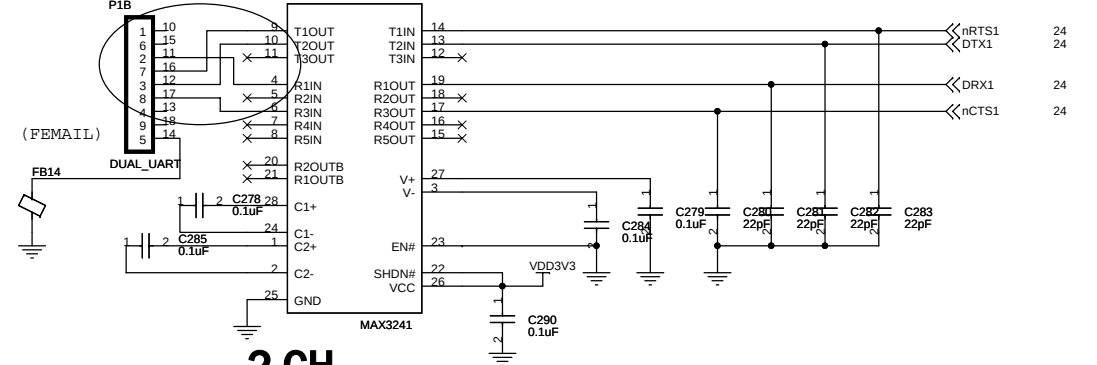
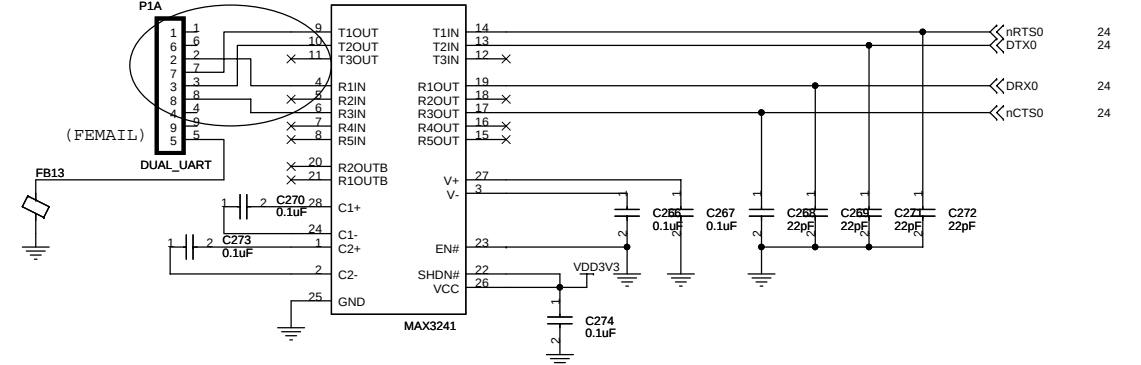
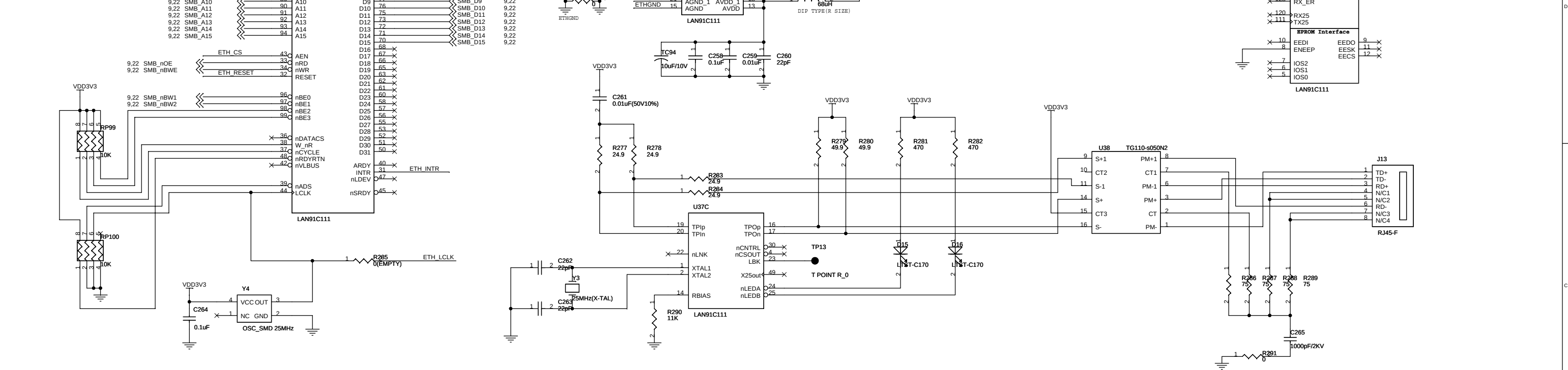
HEADPHONE_OUT



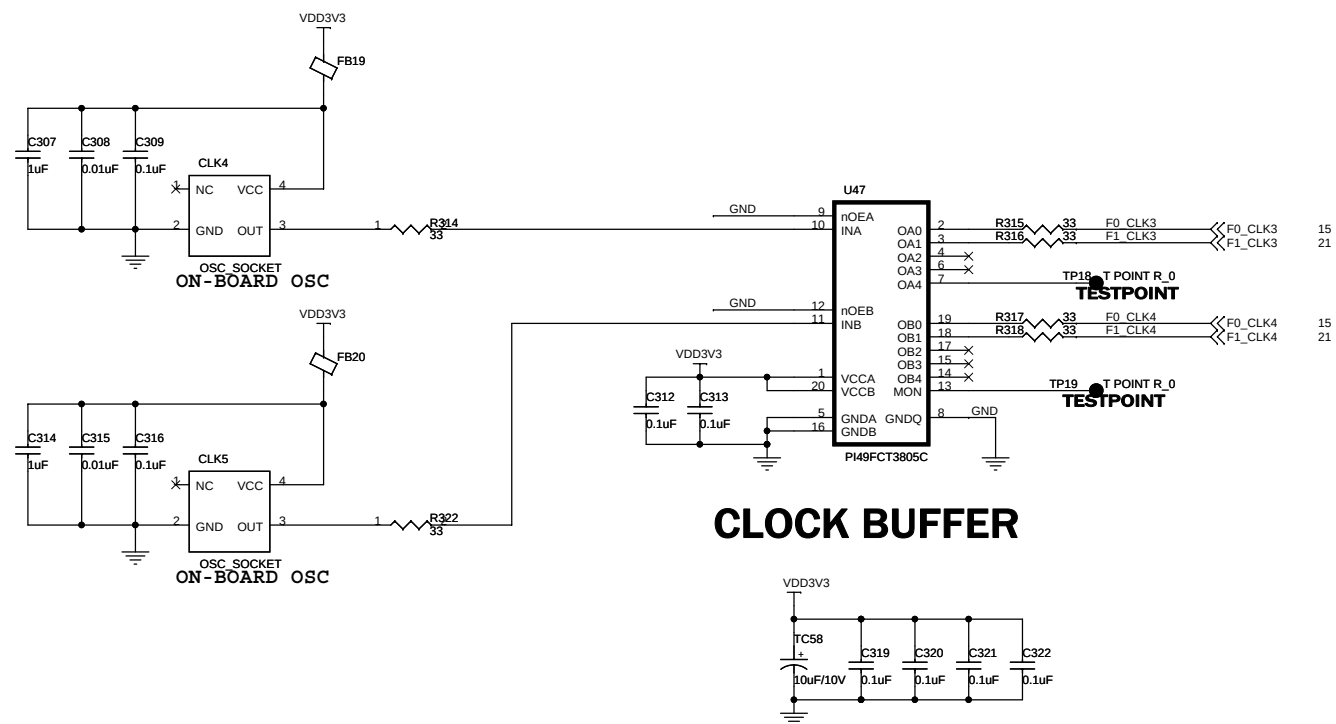
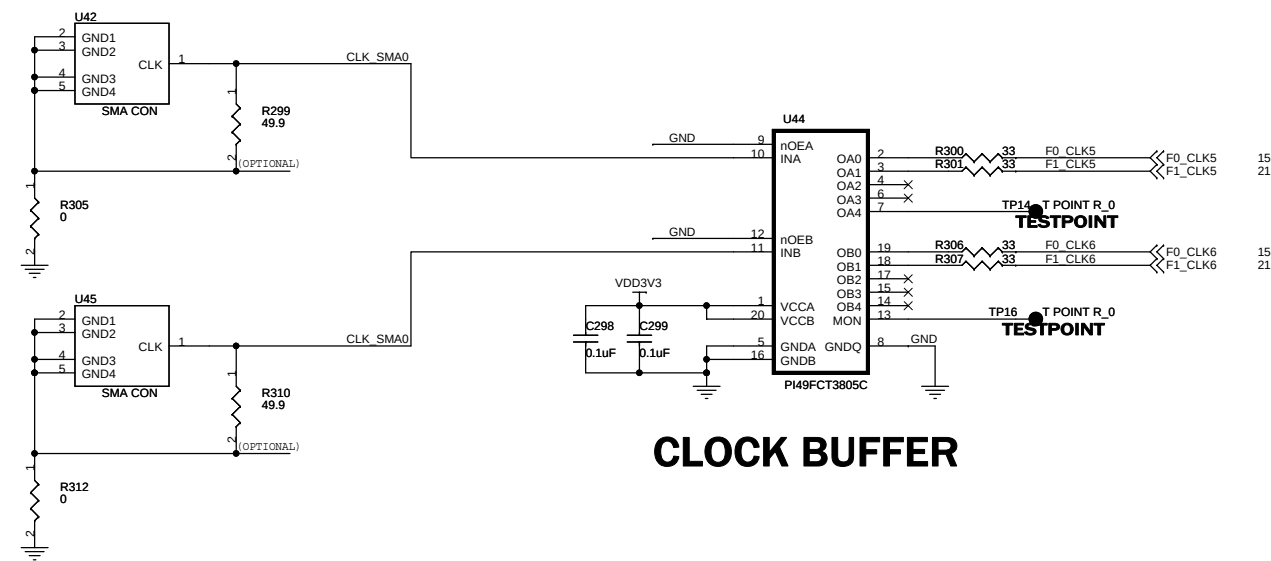
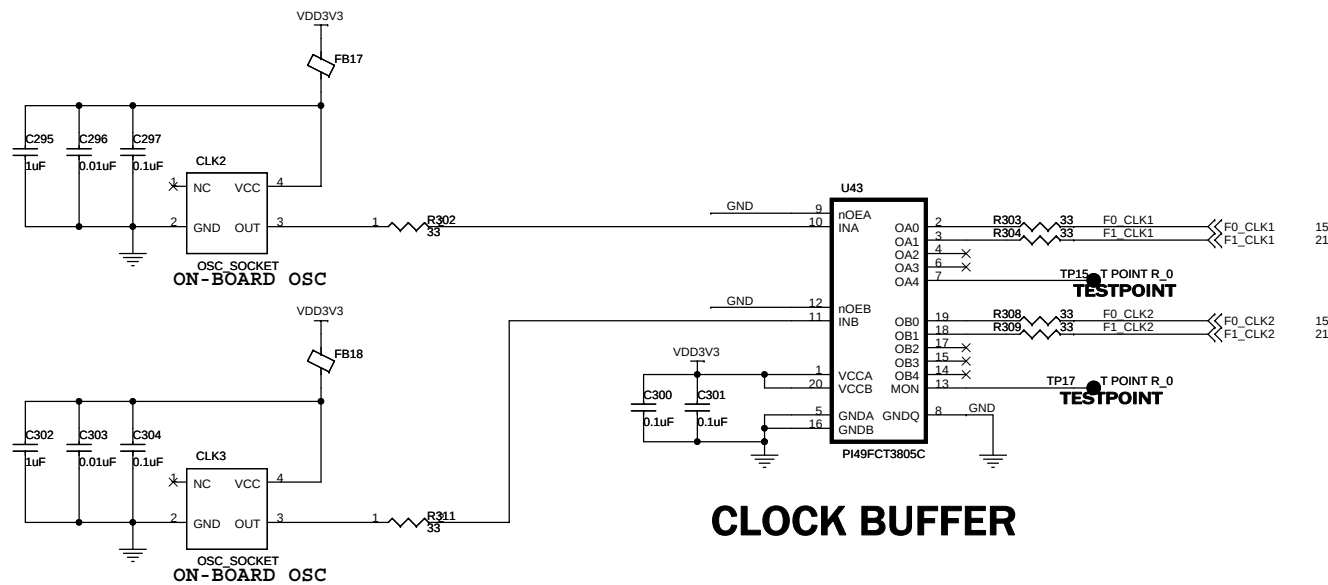
IIS Interface (Audio Codec)

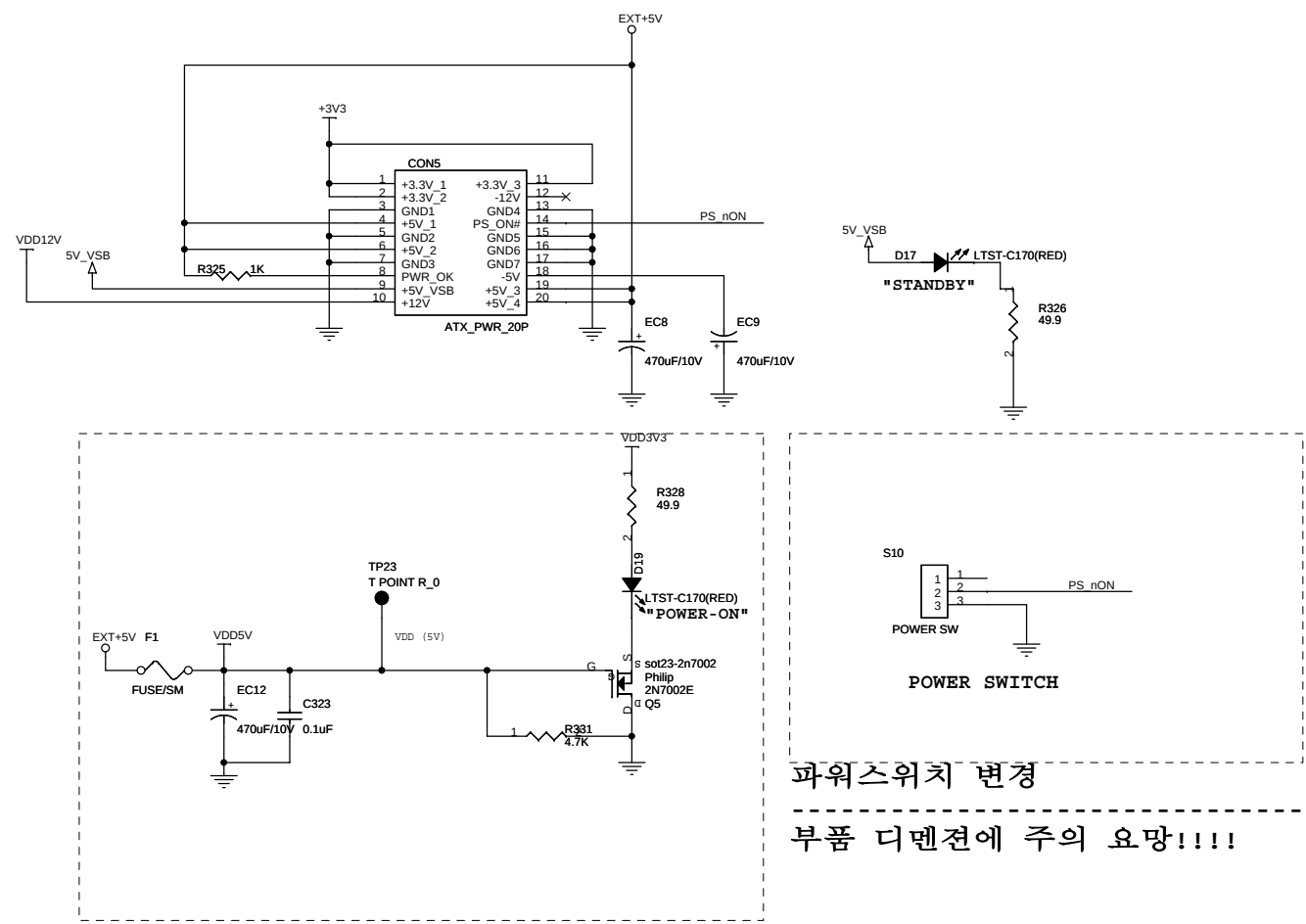
SIGNALS

22	ETH_INTR	ETH_INTR
22	ETH_CS	ETH_CS
22	ETH_RESET	ETH_RESET
22	ETH_LCLK	ETH_LCLK



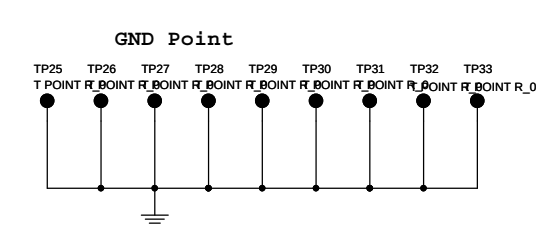
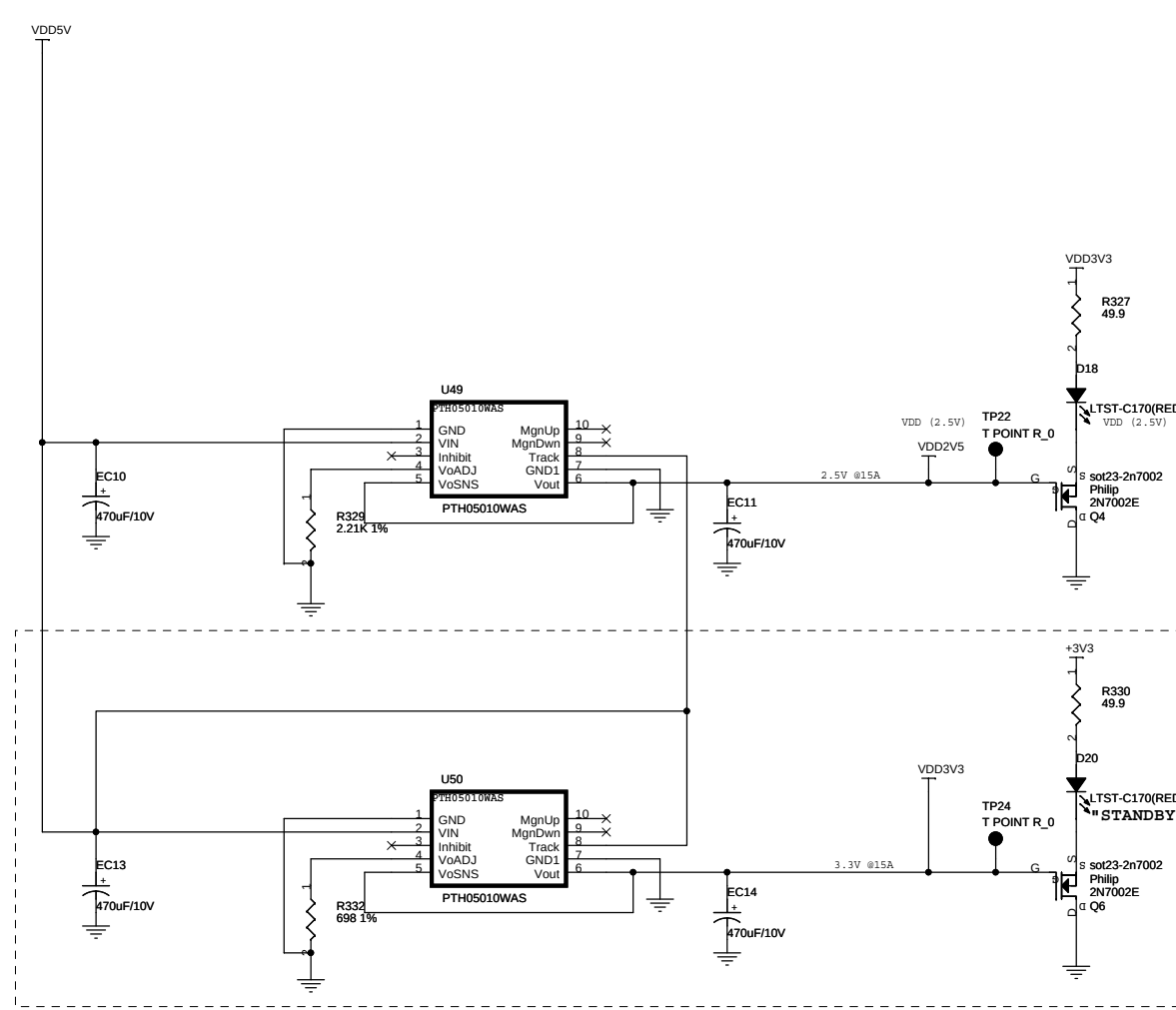
2-CH UART



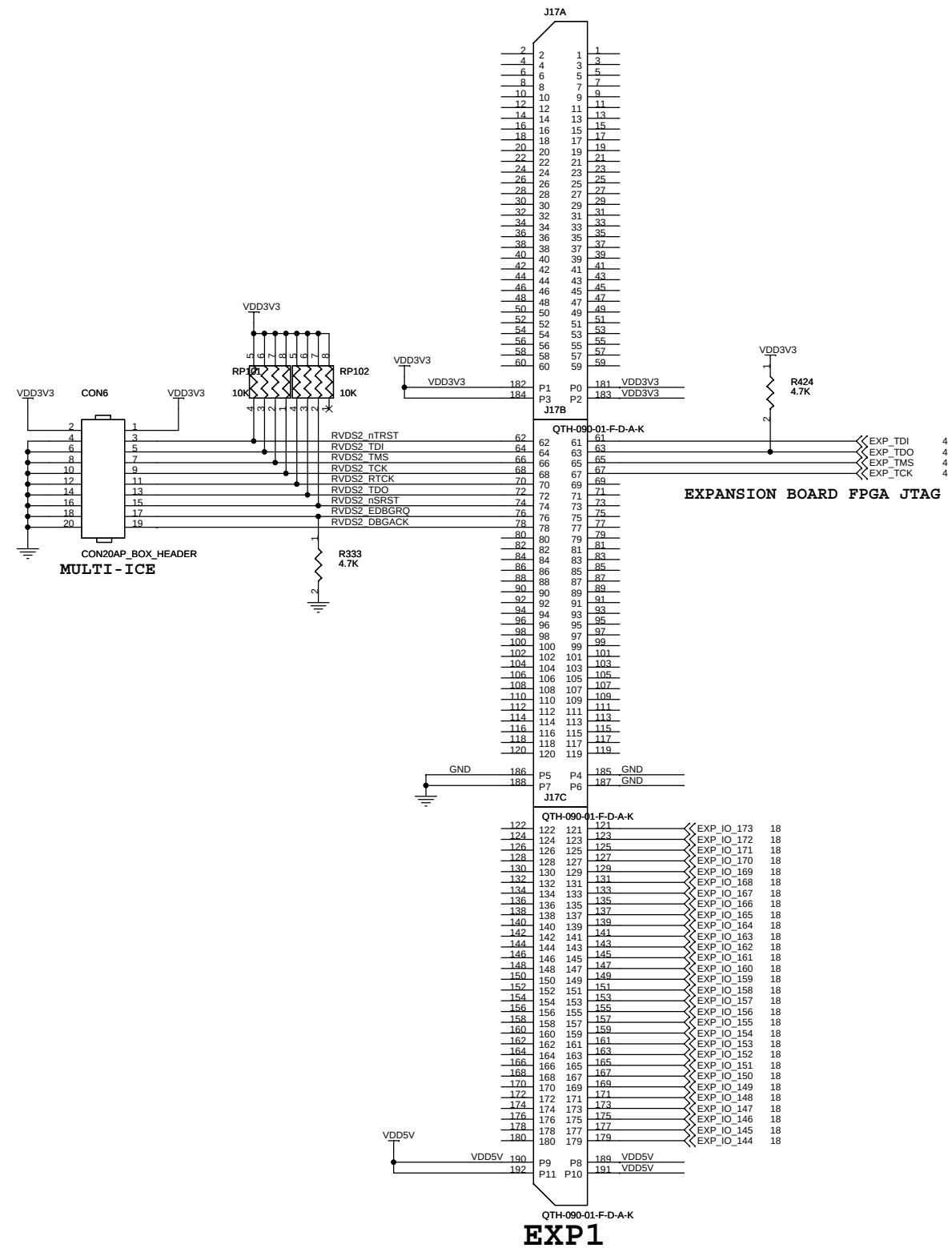
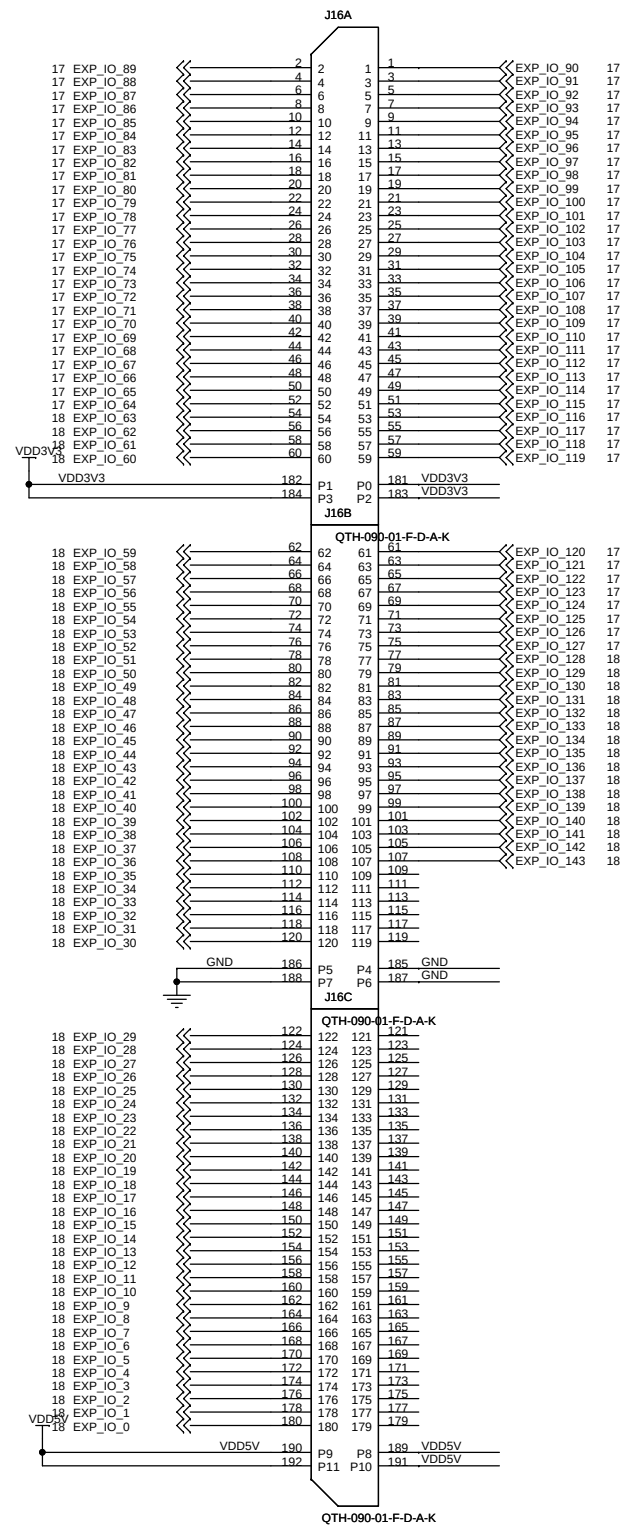


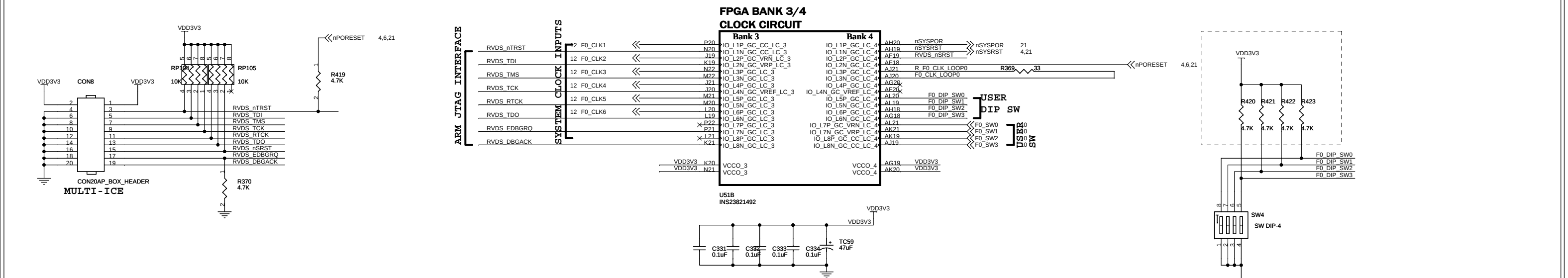
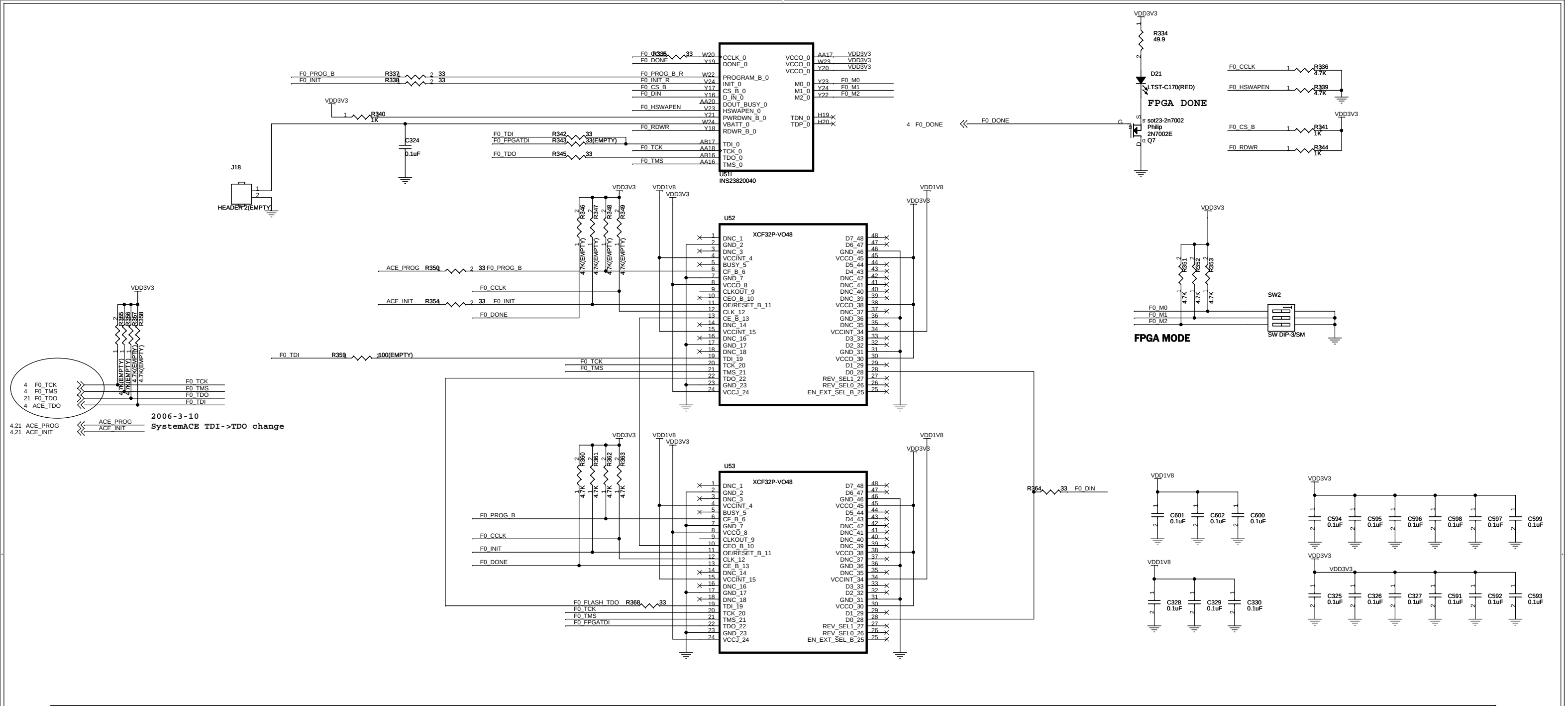
파워스위치 변경

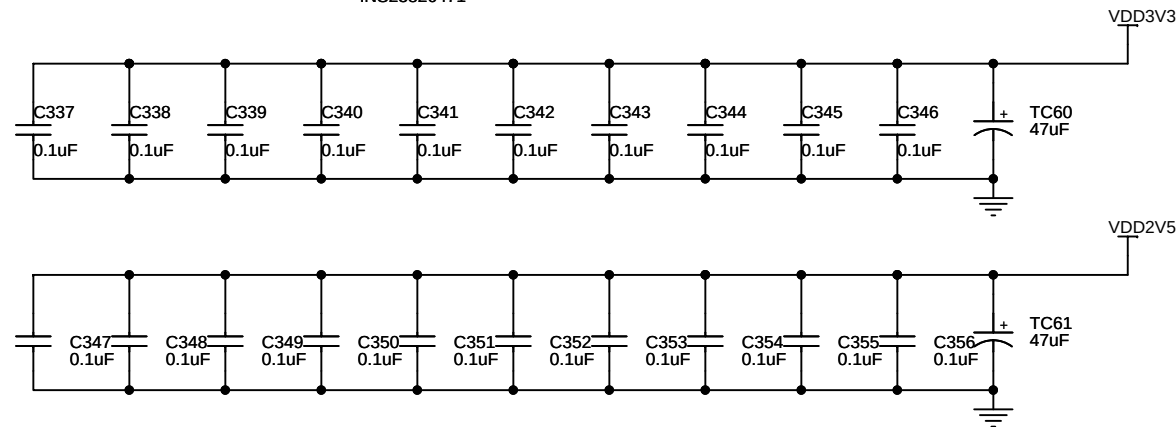
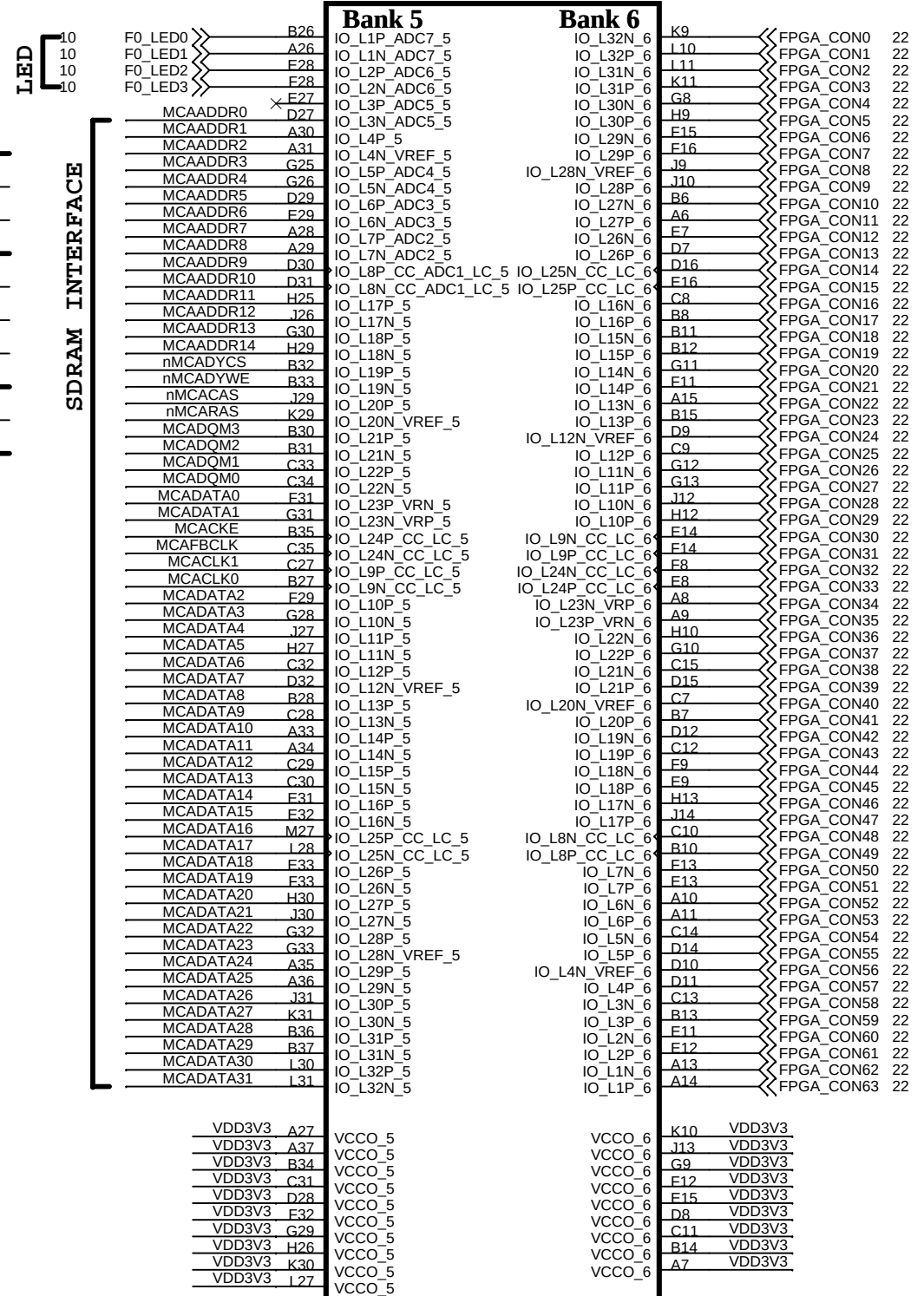
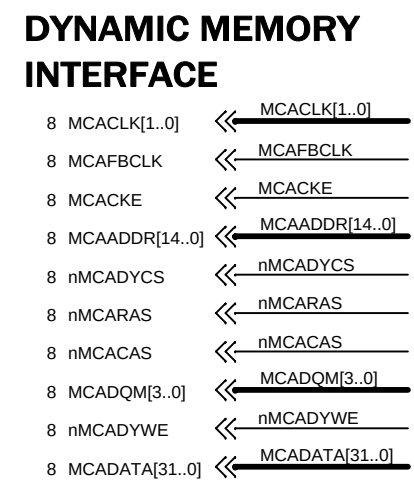
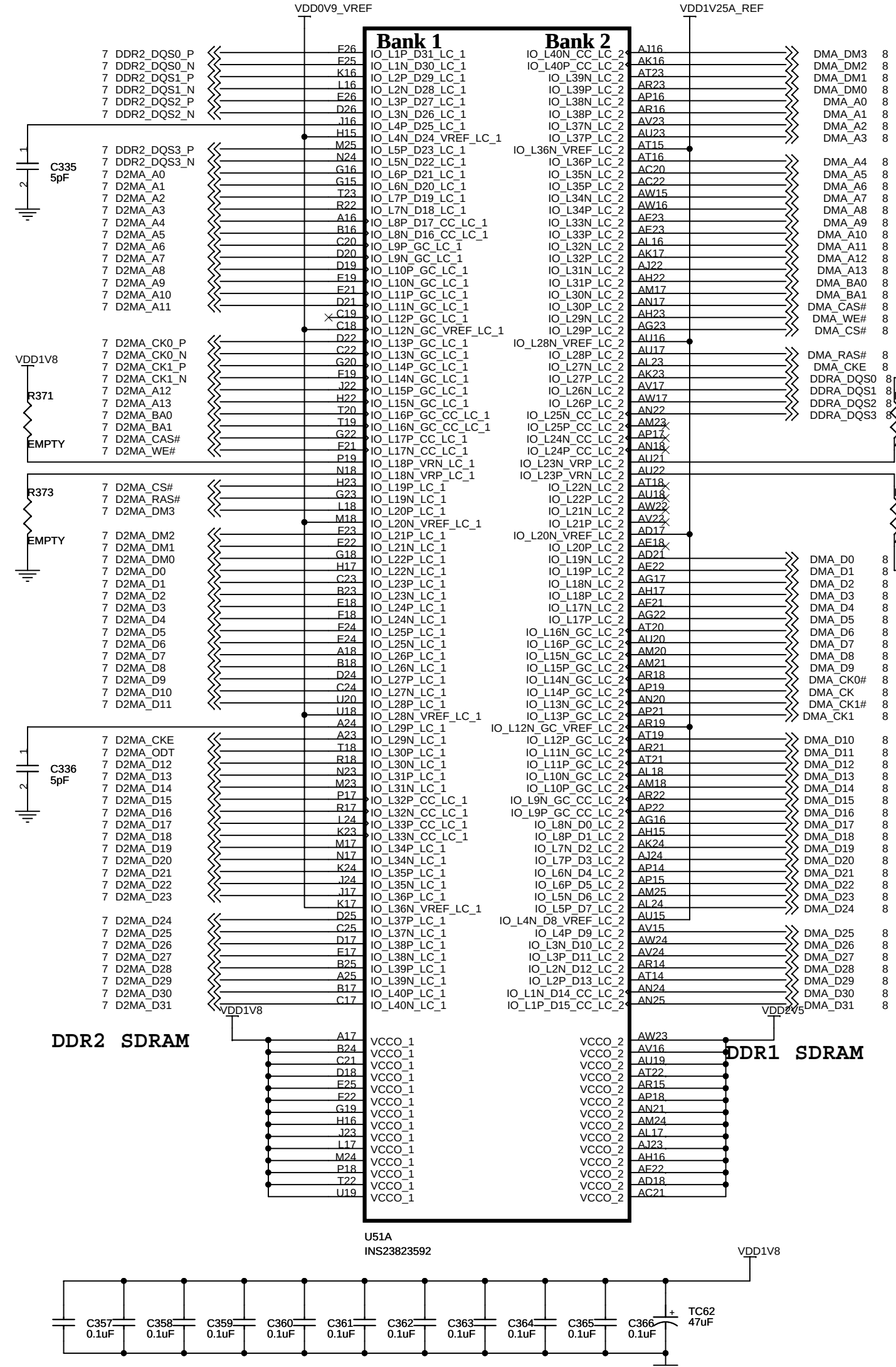
부품 디멘전에 주의 요망!!!!



EXPANSION SLOT FOR FPGA #0

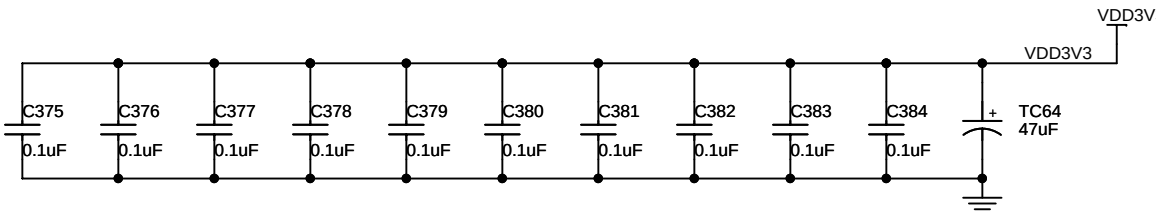
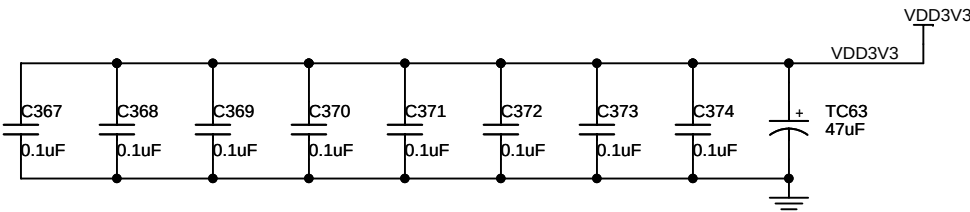






FPGA INTERCONNECTION

Bank 7										Bank 8										Bank 9										Bank 10									
23	FPGA_CON384	AT33	IO_L1P_7	IO_L32N_8	AW10	FPGA_CON320	23	EXP_IO_64	D34	IO_L1P_9	IO_L32N_10	T11	FPGA_CON64	23																									
23	FPGA_CON385	AR33	IO_L1N_7	IO_L32P_8	AW11	FPGA_CON321	23	EXP_IO_65	D35	IO_L1N_9	IO_L32P_10	U12	FPGA_CON65	23																									
23	FPGA_CON386	AJ30	IO_L2P_7	IO_L31N_8	AV14	FPGA_CON322	23	EXP_IO_66	C37	IO_L2P_9	IO_L31N_10	R8	FPGA_CON66	23																									
23	FPGA_CON387	AK31	IO_L2N_7	IO_L31P_8	AW14	FPGA_CON323	23	EXP_IO_67	C38	IO_L2N_9	IO_L31P_10	R9	FPGA_CON67	23																									
23	FPGA_CON388	AM30	IO_L3P_7	IO_L30N_8	AU10	FPGA_CON324	23	EXP_IO_68	F34	IO_L3P_9	IO_L30N_10	P7	FPGA_CON68	23																									
23	FPGA_CON389	AL30	IO_L3N_7	IO_L30P_8	AV10	FPGA_CON325	23	EXP_IO_69	F34	IO_L3N_9	IO_L30P_10	N7	FPGA_CON69	23																									
23	FPGA_CON390	AM31	IO_L4P_7	IO_L29N_8	AR13	FPGA_CON326	23	EXP_IO_70	E35	IO_L4P_9	IO_L29N_10	M6	FPGA_CON70	23																									
23	FPGA_CON391	AL31	IO_L4N_VREF_7	IO_L29P_8	AT13	FPGA_CON327	23	EXP_IO_71	G35	IO_L4N_VREF_9	IO_L29P_10	L6	FPGA_CON71	23																									
23	FPGA_CON392	AP24	IO_L5P_7	IO_L28N_VREF_8	AR11	FPGA_CON328	23	EXP_IO_72	D36	IO_L5P_9	IO_L28N_VREF_10	N8	FPGA_CON72	23																									
23	FPGA_CON393	AR24	IO_L5N_7	IO_L28P_8	AT11	FPGA_CON329	23	EXP_IO_73	D37	IO_L5N_9	IO_L28P_10	P9	FPGA_CON73	23																									
23	FPGA_CON394	AP32	IO_L6P_7	IO_L27N_8	AM13	FPGA_CON330	23	EXP_IO_74	H33	IO_L6P_9	IO_L27N_10	M7	FPGA_CON74	23																									
23	FPGA_CON395	AN32	IO_L6N_7	IO_L27P_8	AN14	FPGA_CON331	23	EXP_IO_75	H34	IO_L6N_9	IO_L27P_10	M8	FPGA_CON75	23																									
23	FPGA_CON396	AN30	IO_L7P_7	IO_L26N_8	AV9	FPGA_CON332	23	EXP_IO_76	E36	IO_L7P_9	IO_L26N_10	U13	FPGA_CON76	23																									
23	FPGA_CON397	AP31	IO_L7N_7	IO_L26P_8	AW9	FPGA_CON333	23	EXP_IO_77	F36	IO_L7N_9	IO_L26P_10	T13	FPGA_CON77	23																									
23	FPGA_CON398	AK29	IO_L8P_CC_LC_7	IO_L25N_CC_LC_8	AV12	FPGA_CON334	23	EXP_IO_78	C39	IO_L8P_CC_LC_9	IO_L25N_CC_LC_10	R11	FPGA_CON78	23																									
23	FPGA_CON399	AJ29	IO_L8N_CC_LC_7	IO_L25P_CC_LC_8	AW12	FPGA_CON335	23	EXP_IO_79	D39	IO_L8N_CC_LC_9	IO_L25P_CC_LC_10	P11	FPGA_CON79	23																									
23	FPGA_CON400	AT24	IO_L9P_CC_LC_7	IO_L24N_CC_LC_8	AV7	FPGA_CON336	23	EXP_IO_80	J32	IO_L9P_CC_LC_9	IO_L24N_CC_LC_10	G1	FPGA_CON80	23																									
23	FPGA_CON401	AT25	IO_L9N_CC_LC_7	IO_L24P_CC_LC_8	AV8	FPGA_CON337	23	EXP_IO_81	K32	IO_L9N_CC_LC_9	IO_L24P_CC_LC_10	F1	FPGA_CON81	23																									
23	FPGA_CON402	AW34	IO_L10P_7	IO_L23N_VRP_8	AU11	FPGA_CON338	23	EXP_IO_82	G36	IO_L10P_9	IO_L23N_VRP_10	K6	FPGA_CON82	23																									
23	FPGA_CON403	AV34	IO_L10N_7	IO_L23P_VRN_8	AU12	FPGA_CON339	23	EXP_IO_83	G37	IO_L10N_9	IO_L23P_VRN_10	J6	FPGA_CON83	23																									
23	FPGA_CON404	AW30	IO_L11P_7	IO_L22N_8	AT9	FPGA_CON340	23	EXP_IO_84	E37	IO_L11P_9	IO_L22N_10	G2	FPGA_CON84	23																									
23	FPGA_CON405	AW31	IO_L11N_7	IO_L22P_8	AT10	FPGA_CON341	23	EXP_IO_85	F38	IO_L11N_9	IO_L22P_10	G3	FPGA_CON85	23																									
23	FPGA_CON406	AV33	IO_L12P_7	IO_L21N_8	AU13	FPGA_CON342	23	EXP_IO_86	H35	IO_L12P_9	IO_L21N_10	J5	FPGA_CON86	23																									
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23	FPGA_CON409	AP26	IO_L13N_7	IO_L20P_8	AR9	FPGA_CON345	23	EXP_IO_89	M31	IO_L13N_9	IO_L20P_10	K7	FPGA_CON89	23																									
23	FPGA_CON410	AT31	IO_L14P_7	IO_L19N_8	AP12	FPGA_CON346	23	EXP_IO_90	F39	IO_L14P_9	IO_L19N_10	R12	FPGA_CON90	23																									
23	FPGA_CON411	AT30	IO_L14N_7	IO_L19P_8	AR12	FPGA_CON347	23	EXP_IO_91	F39	IO_L14N_9	IO_L19P_10	P12	FPGA_CON91	23																									
23	FPGA_CON412	AU25	IO_L15P_7	IO_L18N_8	AT8	FPGA_CON348	23	EXP_IO_92	J34	IO_L15P_9	IO_L18N_10	E1	FPGA_CON92	23																									
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23	FPGA_CON414	AR31	IO_L16P_7	IO_L17N_8	AL13	FPGA_CON350	23	EXP_IO_94	E38	IO_L16P_9	IO_L17N_10	T14	FPGA_CON94	23																									
23	FPGA_CON415	AR32	IO_L16N_7	IO_L17P_8	AL14	FPGA_CON351	23	EXP_IO_95	G38	IO_L16N_9	IO_L17P_10	R14	FPGA_CON95	23																									
23	FPGA_CON416	AI26	IO_L17P_7	IO_L16N_8	AU6	FPGA_CON352	23	EXP_IO_96	K33	IO_L17P_9	IO_L16N_10	D1	FPGA_CON96	23																									
23	FPGA_CON417	AM27	IO_L17N_7	IO_L16P_8	AU7	FPGA_CON353	23	EXP_IO_97	I33	IO_L17N_9	IO_L16P_10	D2	FPGA_CON97	23																									
23	FPGA_CON418	AU31	IO_L18P_7	IO_L15N_8	AP11	FPGA_CON354	23	EXP_IO_98	H37	IO_L18P_9	IO_L15N_10	F3	FPGA_CON98	23																									
23	FPGA_CON419	AU32	IO_L18N_7	IO_L15P_8	AN12	FPGA_CON355	23	EXP_IO_99	H38	IO_L18N_9	IO_L15P_10	E4	FPGA_CON99	23																									
23	FPGA_CON420	AV28	IO_L19P_7	IO_L14N_8	AJ12	FPGA_CON356	23	EXP_IO_100	N30	IO_L19P_9	IO_L14N_10	N10	FPGA_CON100	23																									
23	FPGA_CON421	AU28	IO_L19N_7	IO_L14P_8	AH13	FPGA_CON357	23	EXP_IO_101	P29	IO_L19N_9	IO_L14P_10	M10	FPGA_CON101	23																									
23	FPGA_CON422	AW32	IO_L20P_7	IO_L13N_8	AN10	FPGA_CON358	23	EXP_IO_102	I34	IO_L20P_9	IO_L13N_10	T15	FPGA_CON102	23																									
23	FPGA_CON423	AV32	IO_L20N_VREF_7	IO_L13P_8	AM11	FPGA_CON359	23	EXP_IO_103	I35	IO_L20N_VREF_9	IO_L13P_10	R16	FPGA_CON103	23																									
23	FPGA_CON424	AW25	IO_L21P_7	IO_L12N_VREF_8	AI10	FPGA_CON360	23	EXP_IO_104	J36	IO_L21P_9	IO_L12N_VREF_10	E3	FPGA_CON104	23																									
23	FPGA_CON425	AW26	IO_L21N_7	IO_L12P_8	AM10	FPGA_CON361	23	EXP_IO_105	J37	IO_L21N_9	IO_L12P_10	F4	FPGA_CON105	23																									
23	FPGA_CON426	AP29	IO_L22P_7	IO_L11N_8	AW6	FPGA_CON362	23	EXP_IO_106	K36	IO_L22P_9	IO_L11N_10	N12	FPGA_CON106	23																									
23	FPGA_CON427	AN29	IO_L22N_7	IO_L11P_8	AW7	FPGA_CON363	23	EXP_IO_107	I36	IO_L22N_9	IO_L11P_10	M11	FPGA_CON107	23																									
23	FPGA_CON428	AN27	IO_L23P_VRN_7	IO_L10N_8	AW4	FPGA_CON364	23	EXP_IO_108	H39	IO_L23P_VRN_9	IO_L10N_10	G5	FPGA_CON108	23																									
23	FPGA_CON429	AN28	IO_L23N_VRP_7	IO_L10P_8	AW5	FPGA_CON365	23	EXP_IO_109	J39	IO_L23N_VRP_9	IO_L10P_10	E5	FPGA_CON109	23																									
23	FPGA_CON430	AL28	IO_L24P_CC_LC_7	IO_L9N_CC_LC_8	AK11	FPGA_CON366	23	EXP_IO_110	M33	IO_L24P_CC_LC_9	IO_L9N_CC_LC_10	G6	FPGA_CON110	23																									
23	FPGA_CON431	AM28	IO_L24N_CC_LC_7	IO_L9P_CC_LC_8	AL11	FPGA_CON367	23	EXP_IO_111	N32	IO_L24N_CC_LC_9	IO_L9P_CC_LC_10	G7	FPGA_CON111	23																									
23	FPGA_CON432	AP27	IO_L25P_CC_SM7_LC_7	IO_L8N_CC_LC_8	AV3	FPGA_CON368	23	EXP_IO_112	R28	IO_L25P_CC_LC_9	IO_L8N_CC_LC_10	J7	FPGA_CON112	23																									
23	FPGA_CON433	AR27	IO_L25N_CC_SM7_LC_7	IO_L8P_CC_LC_8	AV4	FPGA_CON369	23	EXP_IO_113	R29	IO_L25N_CC_LC_9	IO_L8P_CC_LC_10	H7	FPGA_CON113	23																									
23	FPGA_CON434	AV30	IO_L26P_SM6_7	IO_L7N_8	AR7	FPGA_CON370	23	EXP_IO_114	M35	IO_L26P_9	IO_L7N_10	F6	FPGA_CON114	23																									
23	FPGA_CON435	AU30	IO_L26N_SM6_7	IO_L7P_8	AR8	FPGA_CON371	23	EXP_IO_115	N35	IO_L26N_9	IO_L7P_10	F6	FPGA_CON115	23																									
23	FPGA_CON436	AR26	IO_L27P_SM5_7	IO_L6N_8	AN7	FPGA_CON372	23	EXP_IO_116	K37	IO_L27P_9	IO_L6N_10	C2	FPGA_CON116	23																									
23	FPGA_CON437	AT26	IO_L27N_SM5_7	IO_L6P_8	AP7	FPGA_CON373	23	EXP_IO_117	K38	IO_L27N_9	IO_L6P_10	C3	FPGA_CON117	23																									
23	FPGA_CON438	AW29	IO_L28P_7	IO_L5N_8	AN15	FPGA_CON374	23	EXP_IO_118	N33	IO_L28P_9	IO_L5N_10	D5	FPGA_CON118	23																									
23	FPGA_CON439	AV29	IO_L28N_VREF_7	IO_L5P_8	AM15	FPGA_CON375	23	EXP_IO_119	N34	IO_L28N_VREF_9	IO_L5P_10	D6	FPGA_CON119	23																									
23	FPGA_CON440	AV27	IO_L29P_SM4_7	IO_L4N_VREF_8	AK9	FPGA_CON376	23	EXP_IO_120	P31	IO_L29P_9	IO_L4N_VREF_10	D4	FPGA_CON120	23																									
23	FPGA_CON441	AW27	IO_L29N_SM4_7	IO_L4P_8	AL9	FPGA_CON377	23	EXP_IO_121	P32	IO_L29N_9	IO_L4P_10	C4	FPGA_CON121	23																									
23	FPGA_CON442	AT29	IO_L30P_SM3_7	IO_L3N_8	AN8	FPGA_CON378	23	EXP_IO_122	K39	IO_L30P_9	IO_L3N_10	C5	FPGA_CON122	23																									
23	FPGA_CON443	AR29	IO_L30N_SM3_7	IO_L3P_8	AN9	FPGA_CON379	23	EXP_IO_123	I39	IO_L30N_9	IO_L3P_10	B5	FPGA_CON123	23																									
23	FPGA_CON444	AJ26	IO_L31P_SM2_7	IO_L2N_8	AJ9	FPGA_CON380	23	EXP_IO_124	I38	IO_L31P_9	IO_L2N_10	B3	FPGA_CON124	23																									
23	FPGA_CON445	AU27	IO_L31N_SM2_7	IO_L2P_8	AJ10	FPGA_CON381	23	EXP_IO_125	M38	IO_L31N_9	IO_L2P_10	A3	FPGA_CON125	23																									
23	FPGA_CON446	AT28	IO_L32P_SM1_7	IO_L1N_8	AJ5	FPGA_CON382	23	EXP_IO_126	M36	IO_L32P_9	IO_L1N_10	A4	FPGA_CON126	23																									
23	FPGA_CON447	AR28	IO_L32N_SM1_7	IO_L1P_8	AV5	FPGA_CON383	23	EXP_IO_127	M37	IO_L32N_9	IO_L1P_10	A5	FPGA_CON127	23																									



8 SMA_NOE

SSRAM INTERFACE

NAND FLASH

8 NDAREADY
8 NDARE
8 NDACE
8 NDACLE
8 NDAALE
8 NDAAnWE
8 NDA_D7
8 NDA_D6
8 NDA_D5
8 NDA_D4
8 NDA_D3
8 NDA_D2
8 NDA_D1
8 NDA_D0

14 EXP_IO_173
14 EXP_IO_172
14 EXP_IO_171
14 EXP_IO_169
14 EXP_IO_168
14 EXP_IO_167
14 EXP_IO_166
14 EXP_IO_165
14 EXP_IO_164
14 EXP_IO_163
14 EXP_IO_162
14 EXP_IO_161
14 EXP_IO_160
14 EXP_IO_159
14 EXP_IO_158
14 EXP_IO_157
14 EXP_IO_156
14 EXP_IO_155
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14 EXP_IO_152
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14 EXP_IO_136
14 EXP_IO_135
14 EXP_IO_134
14 EXP_IO_133
14 EXP_IO_132
14 EXP_IO_131
14 EXP_IO_130
14 EXP_IO_129
14 EXP_IO_128

VDD3V3 AD28
VDD3V3 AE25
VDD3V3 AF32
VDD3V3 AG29
VDD3V3 AI33
VDD3V3 AM34
VDD3V3 AP38
VDD3V3 AR35
VDD3V3 AU39
VDD3V3 AV36

VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11
VCCO_11

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Bank 11

IO_L1P_11
IO_L1N_11
IO_L2P_11
IO_L2N_11
IO_L3P_11
IO_L3N_11
IO_L4P_11
IO_L4N_VREF_11
IO_L5P_11
IO_L5N_11
IO_L6P_11
IO_L6N_11
IO_L7P_11
IO_L7N_11
IO_L8P_CC_LC_11
IO_L8N_CC_LC_11
IO_L9P_CC_LC_11
IO_L9N_CC_LC_11
IO_L10P_11
IO_L10N_11
IO_L11P_11
IO_L11N_11
IO_L12P_11
IO_L12N_VREF_11
IO_L13P_11
IO_L13N_11
IO_L14P_11
IO_L14N_11
IO_L15P_11
IO_L15N_11
IO_L16P_11
IO_L16N_11
IO_L17P_11
IO_L17N_11
IO_L18P_11
IO_L18N_11
IO_L19P_11
IO_L19N_11
IO_L20P_11
IO_L20N_VREF_11
IO_L21P_11
IO_L21N_11
IO_L22P_11
IO_L22N_11
IO_L23P_VRN_11
IO_L23N_VRP_11
IO_L24P_CC_LC_11
IO_L24N_CC_LC_11
IO_L25P_CC_LC_11
IO_L25N_CC_LC_11
IO_L26P_11
IO_L26N_11
IO_L27P_11
IO_L27N_11
IO_L28P_11
IO_L28N_VREF_11
IO_L29P_11
IO_L29N_11
IO_L30P_11
IO_L30N_11
IO_L31P_11
IO_L31N_11
IO_L32P_11
IO_L32N_11

VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12
VCCO_12

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FPGA INTERCONNECTION

AT3
AU3
AT5
AT6
AP4
AP5
AP6
AR6
AM6
AM7
AR4
AT4
AR2
AR3
AU1
AU2
AR1
AT1
AK8
AL8
AL5
AM5
AN4
AN5
AJ7
AK7
AH9
AH10
AN2
AN3
AK6
AL6
AK4
AL4
AP1
AP2
AM1
AM2
AL3
AM3
AE11
AE12
AG10
AF11
AK2
AK3
AJ5
AJ6
AH4
AJ4
AH7
AG8
AF8
AF9
AK1
AJ1
AJ2
AG6
AG7
AH2
AH3
AG5
AH5

AT2
AR5
AN1
AM4
AL7
AJ3
AH6
AG9
AF12

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Bank 13

IO_L1P_13
IO_L1N_13
IO_L2P_13
IO_L2N_13
IO_L3P_13
IO_L3N_13
IO_L4P_13
IO_L4N_VREF_13
IO_L5P_13
IO_L5N_13
IO_L6P_13
IO_L6N_13
IO_L7P_13
IO_L7N_13
IO_L8P_CC_LC_13
IO_L8N_CC_LC_13
IO_L9P_CC_LC_13
IO_L9N_CC_LC_13
IO_L10P_13
IO_L10N_13
IO_L11P_13
IO_L11N_13
IO_L12P_13
IO_L12N_VREF_13
IO_L13P_13
IO_L13N_13
IO_L14P_13
IO_L14N_13
IO_L15P_13
IO_L15N_13
IO_L16P_13
IO_L16N_13
IO_L17P_13
IO_L17N_13
IO_L18P_13
IO_L18N_13
IO_L19P_13
IO_L19N_13
IO_L20P_13
IO_L20N_VREF_13
IO_L21P_13
IO_L21N_13
IO_L22P_13
IO_L22N_13
IO_L23P_VRN_13
IO_L23N_VRP_13
IO_L24P_CC_LC_13
IO_L24N_CC_LC_13
IO_L25P_CC_LC_13
IO_L25N_CC_LC_13
IO_L26P_13
IO_L26N_13
IO_L27P_13
IO_L27N_13
IO_L28P_13
IO_L28N_VREF_13
IO_L29P_13
IO_L29N_13
IO_L30P_13
IO_L30N_13
IO_L31P_13
IO_L31N_13
IO_L32P_13
IO_L32N_13

VDD3V3 AA37
VDD3V3 P38
VDD3V3 R35
VDD3V3 T32
VDD3V3 U29
VDD3V3 U39
VDD3V3 V26
VDD3V3 V36
VDD3V3 W33

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Bank 14

IO_L32N_14
IO_L32P_14
IO_L31N_14
IO_L31P_14
IO_L30N_14
IO_L30P_14
IO_L29N_14
IO_L29P_14
IO_L28N_VREF_14
IO_L28P_14
IO_L27N_14
IO_L27P_14
IO_L26N_14
IO_L26P_14
IO_L25N_CC_LC_14
IO_L25P_CC_LC_14
IO_L24N_CC_LC_14
IO_L24P_CC_LC_14
IO_L23N_VRP_14
IO_L23P_VRN_14
IO_L22N_14
IO_L22P_14
IO_L21N_14
IO_L21P_14
IO_L20N_VREF_14
IO_L20P_14
IO_L19N_14
IO_L19P_14
IO_L18N_14
IO_L18P_14
IO_L17N_14
IO_L17P_14
IO_L16N_14
IO_L16P_14
IO_L15N_14
IO_L15P_14
IO_L14N_14
IO_L14P_14
IO_L13N_14
IO_L13P_14
IO_L12N_VREF_14
IO_L12P_14
IO_L11N_14
IO_L11P_14
IO_L10N_14
IO_L10P_14
IO_L9N_CC_LC_14
IO_L9P_CC_LC_14
IO_L8N_CC_LC_14
IO_L8P_CC_LC_14
IO_L7N_14
IO_L7P_14
IO_L6N_14
IO_L6P_14
IO_L5N_14
IO_L5P_14
IO_L4N_VREF_14
IO_L4P_14
IO_L3N_14
IO_L3P_14
IO_L2N_14
IO_L2P_14
IO_L1N_14
IO_L1P_14

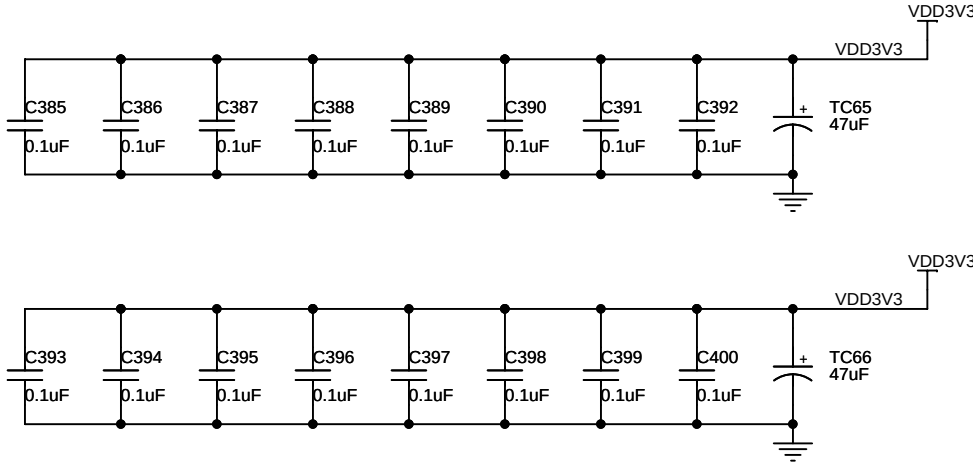
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VCCO_14
VCCO_14
VCCO_14
VCCO_14
VCCO_14
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VCCO_14
VCCO_14
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VCCO_14
VCCO_14
VCCO_14
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VCCO_14
VCCO_14
VCCO_14

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W12
Y13
Y11
AA11
Y9
AA10
W5
W6
W9
W10
W7
V7
U2
U3
U1
T1
V5
U5
V9
V10
U6
U7
T3
T4
R1
R2
T5
T6
R3
R4
T9
U10
P1
P2
U8
T8
V12
V13
N2
N3
P4
P5
R6
P6
M1
L1
N4
N5
M2
M3
L3
L4
K1
K2
M5
L5
J1
J2
H2
H3
K3
K4
J4
H4

Y10
W13
V6
U9
T2
R5
N1
M4
J3

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VIRTEX-4 FPGA
POWER GROUND

GND D13	GND	GND J28	GND
GND K13	GND	GND N28	GND
GND M13	GND	GND W28	GND
GND P13	GND	GND AJ28	GND
GND AD13	GND	GND AW28	GND
GND AE13	GND	GND B29	GND
GND AK13	GND	GND M29	GND
GND AP13	GND	GND AB29	GND
GND G14	GND	GND AM29	GND
GND I14	GND	GND E30	GND
GND N14	GND	GND R30	GND
GND U14	GND	GND AE30	GND
GND W14	GND	GND AR30	GND
GND AE14	GND	GND H31	GND
GND AG14	GND	GND V31	GND
GND AJ14	GND	GND AH31	GND
GND AU14	GND	GND AV31	GND
GND K15	GND	GND A32	GND
GND P15	GND	GND L32	GND
GND V15	GND	GND AA32	GND
GND Y15	GND	GND AL32	GND
GND AD15	GND	GND D33	GND
GND AE15	GND	GND P33	GND
GND AK15	GND	GND AD33	GND
GND C16	GND	GND AP33	GND
GND N16	GND	GND G34	GND
GND U16	GND	GND U34	GND
GND W16	GND	GND AG34	GND
GND AC16	GND	GND AU34	GND
GND AF16	GND	GND K35	GND
GND AN16	GND	GND Y35	GND
GND F17	GND	GND AK35	GND
GND T17	GND	GND C36	GND
GND V17	GND	GND N36	GND
GND AF17	GND	GND AC36	GND
GND AT17	GND	GND AN36	GND
GND J18	GND	GND E37	GND
GND W18	GND	GND T37	GND
GND AC18	GND	GND AE37	GND
GND AJ18	GND	GND AT37	GND
GND AW18	GND	GND A38	GND
GND B19	GND	GND J38	GND
GND M19	GND	GND W38	GND
GND V19	GND	GND AJ38	GND
GND AB19	GND	GND AW38	GND
GND AD19	GND	GND B39	GND
GND AM19	GND	GND M39	GND
GND F20	GND	GND AB39	GND
GND R20	GND	GND AM39	GND
		GND AV39	GND
GND AE20	GND	GND B1	GND
GND AR20	GND	GND H1	GND
GND H21	GND	GND V1	GND
GND T21	GND	GND AH1	GND
GND V21	GND	GND AV1	GND
GND AB21	GND	GND A2	GND
GND AH21	GND	GND L2	GND
GND AV21	GND	GND AA2	GND
GND A22	GND	GND AL2	GND
GND I22	GND	GND AW2	GND
GND U22	GND	GND D3	GND
GND AA22	GND	GND P3	GND
GND AL22	GND	GND AD3	GND
GND D23	GND	GND AP3	GND
GND P23	GND	GND G4	GND
GND AB23	GND	GND U4	GND
GND AD23	GND	GND AG4	GND
GND AP23	GND	GND AU4	GND
GND G24	GND	GND K5	GND
GND R24	GND	GND Y5	GND
GND U24	GND	GND AK5	GND
GND AA24	GND	GND C6	GND
GND AC24	GND	GND N6	GND
GND AG24	GND	GND AC6	GND
GND AU24	GND	GND AN6	GND
GND K25	GND	GND F7	GND
GND P25	GND	GND T7	GND
GND T25	GND	GND AE7	GND
GND Y25	GND	GND AT7	GND
GND AB25	GND	GND J8	GND
GND AE25	GND	GND WR	GND
GND AH25	GND	GND AJ8	GND
GND AK25	GND	GND AW8	GND
GND C26	GND	GND B9	GND
GND I26	GND	GND M9	GND
GND N26	GND	GND AB9	GND
GND R26	GND	GND AM9	GND
GND AA26	GND	GND E10	GND
GND AC26	GND	GND R10	GND
GND AG26	GND	GND AE10	GND
GND AJ26	GND	GND AR10	GND
GND AN26	GND	GND H11	GND
GND F27	GND	GND V11	GND
GND K27	GND	GND AH11	GND
GND P27	GND	GND AV11	GND
GND T27	GND	GND A12	GND
GND AE27	GND	GND L12	GND
GND AH27	GND	GND AA12	GND
GND AK27	GND	GND AG12	GND
GND AT27	GND	GND AL12	GND

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SSRAM INTERFACE

8 SMA_A2	Y27
8 SMA_A3	AA28
8 SMA_A4	W29
8 SMA_A5	W30
8 SMA_A6	AA34
8 SMA_A7	AA35
8 SMA_A8	Y29
8 SMA_A9	AA30
8 SMA_A10	AC38
8 SMA_A11	AC39
8 SMA_A12	AA31
8 SMA_A13	Y31
8 SMA_A14	AC35
8 SMA_A15	AB35
8 SMA_A16	AB33
8 SMA_A17	AA33
8 SMA_A18	AC33
8 SMA_A19	AC34
8 SMA_A20	AD37
8 SMA_nADSC	AC37
8 SMA_nADSP	AC32
8 SMA_nADV	AB31
8 SMA_nBW1	AE39
8 SMA_nBW2	AD39
8 SMA_nBW3	AE38
8 SMA_nBW4	AE39
8 SMA_nBWE	AD35
8 SMA_nCE	AD36
8 SMA_CLK	AC30
8 SMA_nCS1	AB30
8 SMA_nGW	AE37
8 SMA_nLBO	AE38
8 SMA_D31	AB27
8 SMA_D30	AB28
8 SMA_D29	AE34
8 SMA_D28	AD34
8 SMA_D27	AD31
8 SMA_D26	AD32
8 SMA_D25	AE36
8 SMA_D24	AE36
8 SMA_D23	AJ39
8 SMA_D22	AH39
8 SMA_D21	AG37
8 SMA_D20	AG38
8 SMA_D19	AH37
8 SMA_D18	AH38
8 SMA_D17	AE34
8 SMA_D16	AE35
8 SMA_D15	AJ36
8 SMA_D14	AJ37
8 SMA_D13	AG35
8 SMA_D12	AG36
8 SMA_D11	AJ35
8 SMA_D10	AH35
8 SMA_D9	AK38
8 SMA_D8	AK39
8 SMA_D7	AM37
8 SMA_D6	AM38
8 SMA_D5	AL38
8 SMA_D4	AL39
8 SMA_D3	AM36
8 SMA_D2	AL36
8 SMA_D1	AK36
8 SMA_D0	AK37

Bank 15

IO_L1P_15	Y27
IO_L1N_15	AA28
IO_L2P_15	W29
IO_L2N_15	W30
IO_L3P_15	AA34
IO_L3N_15	AA35
IO_L4P_15	Y29
IO_L4N_VREF_15	AA30
IO_L5P_15	AC38
IO_L5N_15	AC39
IO_L6P_15	AA31
IO_L6N_15	Y31
IO_L7P_15	AC35
IO_L7N_15	AB35
IO_L8P_CC_LC_15	AB33
IO_L8N_CC_LC_15	AA33
IO_L9P_CC_LC_15	AC33
IO_L9N_CC_LC_15	AC34
IO_L10P_15	AD37
IO_L10N_15	AC37
IO_L11P_15	AC32
IO_L11N_15	AB31
IO_L12P_15	AE39
IO_L12N_VREF_15	AD39
IO_L13P_15	AE38
IO_L13N_15	AE39
IO_L14P_15	AD35
IO_L14N_15	AD36
IO_L15P_15	AC30
IO_L15N_15	AB30
IO_L16P_15	AE37
IO_L16N_15	AE38
IO_L17P_15	AB27
IO_L17N_15	AB28
IO_L18P_15	AE34
IO_L18N_15	AD34
IO_L19P_15	AD31
IO_L19N_15	AD32
IO_L20P_15	AE36
IO_L20N_VREF_15	AE36
IO_L21P_15	AJ39
IO_L21N_15	AH39
IO_L22P_15	AG37
IO_L22N_15	AG38
IO_L23P_VRN_15	AH37
IO_L23N_VRP_15	AH38
IO_L24P_CC_LC_15	AE34
IO_L24N_CC_LC_15	AE35
IO_L25P_CC_LC_15	AJ36
IO_L25N_CC_LC_15	AJ37
IO_L26P_15	AG35
IO_L26N_15	AG36
IO_L27P_15	AJ35
IO_L27N_15	AH35
IO_L28P_15	AK38
IO_L28N_VREF_15	AK39
IO_L29P_15	AM37
IO_L29N_15	AM38
IO_L30P_15	AL38
IO_L30N_15	AL39
IO_L31P_15	AM36
IO_L31N_15	AL36
IO_L32P_15	AK36
IO_L32N_15	AK37

Bank 16

IO_L32N_16	AF3
IO_L32P_16	AG3
IO_L31N_16	AF8
IO_L31P_16	AF9
IO_L30N_16	AG1
IO_L30P_16	AG2
IO_L29N_16	AE5
IO_L29P_16	AE6
IO_L28N_VREF_16	AD6
IO_L28P_16	AE6
IO_L27N_16	AE4
IO_L27P_16	AE4
IO_L26N_16	AE1
IO_L26P_16	AE1
IO_L25N_CC_LC_16	AD11
IO_L25P_CC_LC_16	AC12
IO_L24N_CC_LC_16	AE2
IO_L24P_CC_LC_16	AE3
IO_L23N_VRP_16	AC7
IO_L23P_VRN_16	AD7
IO_L22N_16	AD9
IO_L22P_16	AC10
IO_L21N_16	AC14
IO_L21P_16	AB15
IO_L20N_VREF_16	AC13
IO_L20P_16	AB13
IO_L19N_16	AD4
IO_L19P_16	AD5
IO_L18N_16	AD1
IO_L18P_16	AD2
IO_L17N_16	AB8
IO_L17P_16	AC8
IO_L16N_16	AC4
IO_L16P_16	AC5
IO_L15N_16	AB10
IO_L15P_16	AB11
IO_L14N_16	AA8
IO_L14P_16	AB7
IO_L13N_16	AC2
IO_L13P_16	AC3
IO_L12N_VREF_16	AB1
IO_L12P_16	AB2
IO_L11N_16	AB5
IO_L11P_16	AB6
IO_L10N_16	AA5
IO_L10P_16	AA6
IO_L9N_CC_LC_16	AA13
IO_L9P_CC_LC_16	AA14
IO_L8N_CC_LC_16	AA1
IO_L8P_CC_LC_16	Y1
IO_L7N_16	AB3
IO_L7P_16	AA3
IO_L6N_16	Y2
IO_L6P_16	Y3
IO_L5N_16	AA4
IO_L5P_16	Y4
IO_L4N_VREF_16	Y6
IO_L4P_16	Y7
IO_L3N_16	W1
IO_L3P_16	W2
IO_L2N_16	V2
IO_L2P_16	V3
IO_L1N_16	W4
IO_L1P_16	V4

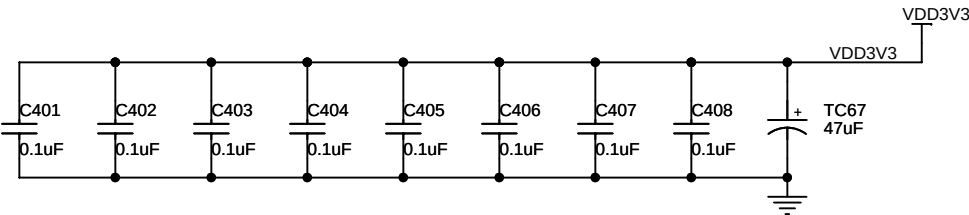
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FPGA_CON197	25
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FPGA_CON200	25
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FPGA_CON202	25
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FPGA_CON219	25
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FPGA_CON254	25
FPGA_CON255	25

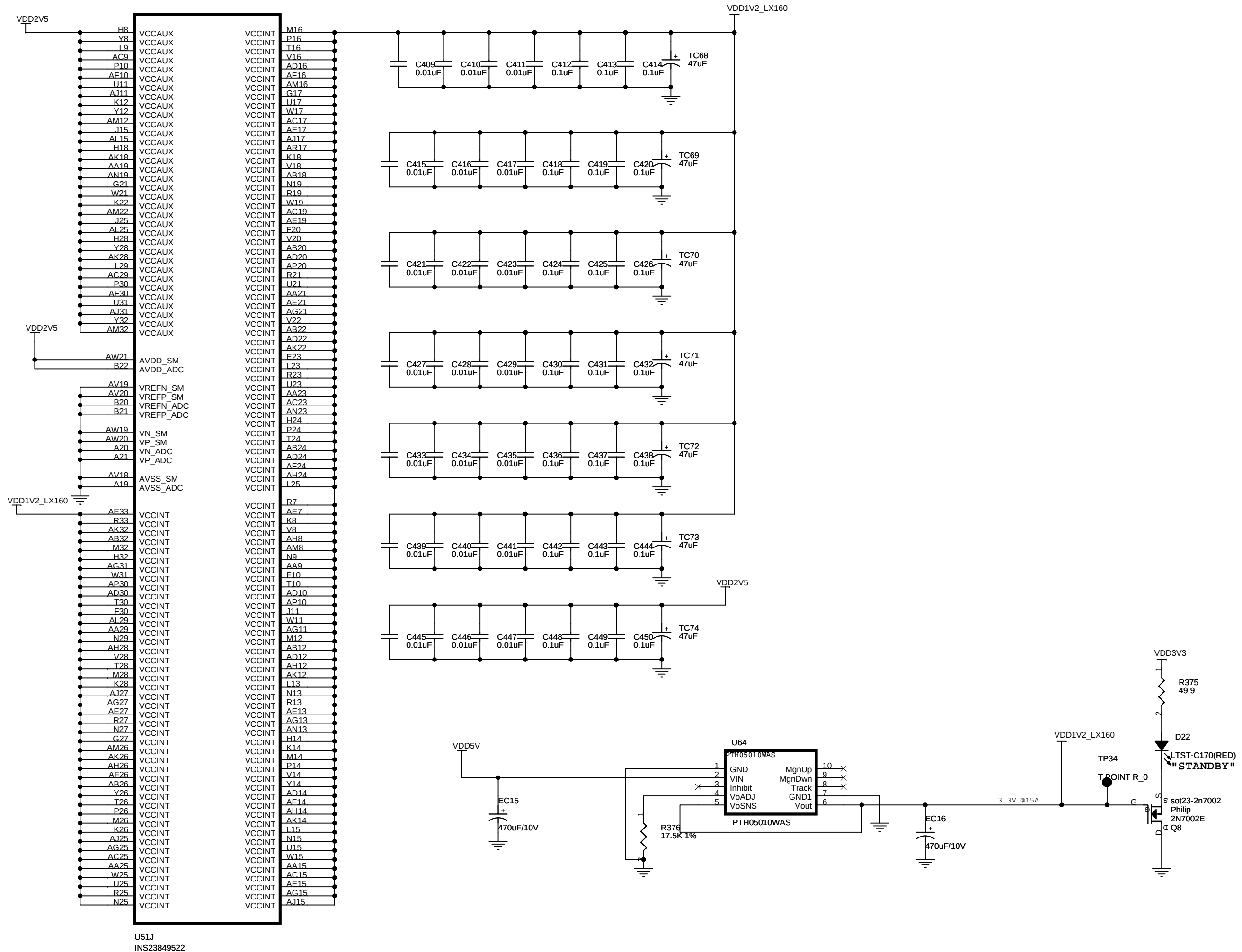
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VDD3V3	AB34
VDD3V3	AC31
VDD3V3	AD38
VDD3V3	AE35
VDD3V3	AG39
VDD3V3	AH36
VDD3V3	AL37
VDD3V3	Y30

VCCO_15	W3
VCCO_15	AE2
VCCO_15	AE5
VCCO_15	AD8
VCCO_15	AC11
VCCO_15	AC1
VCCO_15	AB4
VCCO_15	AB14
VCCO_15	AA7

VDD3V3	W3
VDD3V3	AE2
VDD3V3	AE5
VDD3V3	AD8
VDD3V3	AC11
VDD3V3	AC1
VDD3V3	AB4
VDD3V3	AB14
VDD3V3	AA7

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DYNAMIC MEMORY INTERFACE

